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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 0
Date: 09/04/2021	**By:** A.N

Betterdeck Ltd

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Ireland

Aluminium Profiles Design

Analysis By	Checked By
A.N	T.S.

Revision	Date	Issued By	Comment
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Project: Aluminium Profile	Contract: 1513-1
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Date: 09/04/2021	By: A.N

Contents

Notes: 4

Actions/Colour Code/Result Summary: 4

Actions:….. 4

Colour Code: 4

Result Summary: 4

Aluminium Profile Design For the General Case Scenario (80mm Profile K Series): . 7

Sketch Of Grid Line System For the General Case Scenario For 3kN/m2 Live load and Dead load of Composite Decking (80mm Profile K Series): 7

Aluminium Profile Checked for Live load of 3kN/m2 and Composite Deck for the General Case Scenario (80mm Profile K Series): 8

Sketch Of Grid Line System For the General Case Scenario For 5kN/m2 Live load and Dead load of Composite Decking (80mm Profile K Series): 10

Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Composite Decking for the General Case Scenario (80mm Profile K Series): 11

Aluminium Profile Design For the Occasional Case Scenario (80mm Profile K Series):. 13

Sketch Of Grid Line System For the Occasional Case Scenario for 3kN/m2 Live load and Dead load of Composite Deck (80mm Profile K Series): 13

Aluminium Profile Checked for Live load of 3kN/m2 and Deal load of the composite Decking for the Occasional Case Scenario (80mm Profile K Series): 14

Sketch Of Grid Line System For the Occasional Case Scenario for 5kN/m2 Live load and Dead load of Composite Decking (80mm Profile K Series): 16

Aluminium Profile Checked for Live load of 5kN/m2 and Dead Load of the composite Decking for the Occasional Case Scenario (80mm Profile K Series): 17

Aluminium Profile Design 80mm X 80mm Profile Grid (80mm Profile K Series): 19

Sketch Of Grid Line System For the Last Case Scenario for 3kN/m2 Live load and Dead load of the Composite Decking (80mm Profile K Series): 19

Aluminium Profile Checked for Live load of 3kN/m2 and Composite Deck for the Last Case Scenario (80mm Profile K Series): 20

Sketch Of Grid Line System For the Last Case Scenario for 5kN/m2 Live load and Dead load of the Composite Decking (80mm Profile K Series): 22

Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Composite Decking for the Last Case Scenario (80mm Profile K Series): 23

Aluminium Profile Design For the General Case Scenario (80mm Profile R Series): 25

Sketch Of Grid Line System For the General Case Scenario For 3kN/m2 Live load (80mm Profile R Series): 25

Aluminium Profile Checked for Live load of 3kN/m2 and Dead Load of Pavers for the General Case Scenario (80mm Profile R Series): 26

Sketch Of Grid Line System For the General Case Scenario For 5kN/m2 Live load (80mm Profile R Series): 28

Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Pavers for the General Case Scenario (80mm Profile R Series): 29

Aluminium Profile Design For the Occasional Case Scenario (80mm Profile R Series):. 31

Sketch Of Grid Line System For the Occasional Case Scenario for 3kN/m2 Live load and Dead load of Pavers (80mm Profile R Series): 31

Aluminium Profile Checked for Live load of 3kN/m2 and Dead load of the Pavers the Occasional Case Scenario (80mm Profile R Series): 32

Sketch Of Grid Line System For the Occasional Case Scenario for 5kN/m2 Live load and Dead load of Pavers (80mm Profile R Series): 34

Aluminium Profile Checked for Live load of 5kN/m2 and Dead Load of the Pavers for the Occasional Case Scenario (80mm Profile R Series): 35

Aluminium Profile Design 80mm X 80mm Profile Grid (80mm Profile R Series): 37

Sketch Of Grid Line System For the Last Case Scenario for 3kN/m2 Live load and Dead load of the Pavers (80mm Profile R Series): 37

Aluminium Profile Checked for Live load of 3kN/m2 and Pavers for the Last Case Scenario (80mm Profile R Series): 38

Sketch Of Grid Line System For the Additional Case Scenario for 5kN/m2 Live load and Dead load of the Pavers (80mm Profile R Series): 40

Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Pavers for the Additional Case Scenario (80mm Profile R Series): 41

Aluminium Profile Design For the General Case Scenario (35mm Profile R Series): 43

Sketch Of Grid Line System For the General Case Scenario For 3kN/m2 Live load (35mm Profile R Series): 43

Aluminium Profile Checked for Live load of 3kN/m2 and Dead Load of Pavers for the General Case Scenario (35mm Profile R Series): 44

Sketch Of Grid Line System For the General Case Scenario For 5kN/m2 Live load (35mm Profile R Series): 46

Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Pavers for the General Case Scenario (35mm Profile R Series):.....47

Aluminium Profile Design For the Occasional Case Scenario (35mm Profile R Series):...... 49

Sketch Of Grid Line System For the Occasional Case Scenario for 3kN/m2 Live load and Dead load of the Pavers (35mm Profile R Series): 49

Aluminium Profile Checked for Live load of 3kN/m2 and Deal load of the pavers for the Occasional Case Scenario (35mm Profile R Series): 50

Sketch Of Grid Line System For the Occasional Case Scenario for 5kN/m2 Live load and Dead load of Pavers (35mm Profile R Series): 52

Aluminium Profile Checked for Live load of 5kN/m2 and Dead Load of the Pavers for the Occasional Case Scenario (35mm Profile R Series): 53

Aluminium Profile Design For the General Case Scenario (35mm Profile K Series): 55

Sketch Of Grid Line System For the General Case Scenario For 3kN/m2 Live load (35mm Profile K Series): 55

Aluminium Profile Checked for Live load of 3kN/m2 and Dead Load of Composite Decking for the General Case Scenario (35mm Profile K Series):.....56

Sketch Of Grid Line System For the General Case Scenario For 5kN/m2 Live load (35mm Profile K Series): 58

Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Composite Decking for the General Case Scenario (35mm Profile K Series):.....59

Aluminium Profile Design For the Occasional Case Scenario (35mm Profile K Series):...... 61

Sketch Of Grid Line System For the Occasional Case Scenario for 3kN/m2 Live load and Dead load of the Composite Decking (35mm Profile K Series):.....61

Aluminium Profile Checked for Live load of 3kN/m2 and Deal load of the Composite Decking for the Occasional Case Scenario (35mm Profile K Series):.....62

Sketch Of Grid Line System For the Occasional Case Scenario for 5kN/m2 Live load and Dead load of Composite Decking (35mm Profile K Series):.....64

Aluminium Profile Checked for Live load of 5kN/m2 and Dead Load of the Composite Decking for the Occasional Case Scenario (35mm Profile K Series):.....65

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 4
Date: 09/04/2021	**By:** A.N

Notes:

- The calculations are based on using a span of 400mm between joists supporting only Betterdeck's Recommended Decking and Pavers. The span used for any other flooring material will have to be calculated by that product's manufacturer.
- The spans of 400mm centres for the decking scenarios are for the boards running at 90 degrees to the frame. These spans needs to be 300mm centres should the deck boards run at 45 degrees to the aluminium joists. (see decking and paving manufacture recommendations).
- K Series Profiles are checked for Live loads and Dead load of Composite Deck only as advised by Betterdeck.
- R Series Profiles are checked for Live loads and Dead load of Pavers only as advised by Betterdeck.

Actions/Colour Code for sketches/Result Summary:

Actions:

Live Load $= 3\text{kN/m}^2$ (As per client request)

Live Load $= 5\text{kN/m}^2$ (As per client request)

Colour Code for Sketches:

Yellow Colour → 80mm Profile K-Series.

Green Colour → 80mm Profile R-Series.

Orange Colour → 35mm Profile R-Series.

Pink Colour → 35mm Profile K-Series.

Result Summary:

1- 80mm K Series Profile: 80mm Aluminium Profile Grade A 6063 T-5

1A. The General Case Scenario: (sketch sheets 7 and 10)
Based on a 3 kN/m^2 using composite decking the span between pedestal supports for the 80mm profile will span 1900mm. Based on 5 kN/m^2 using composite decking the 80mm K Series profile will span 1600mm

1B. The Occasional Case Scenario: (sketch sheets 13 and 16)
80mm K Series primary profile and 35mm secondary profile. Pavers not recommended in this case and checks are based on the Live loads and dead load of composite decking. Limiting factors for spacing of secondary 35mm profiles are 900mm for 3 kN/m^2 and 600mm for 5 kN/m^2 as advised by Betterdeck.
Max span between pedestal supports for the 80mm profile is 1500mm for 3 kN/m^2 and 1500mm for the 5kN/m^2.

1C. The Additional Case Scenario: Aluminium Profile Design 80mm X 80mm K Series Profile Grid: (sketch sheets 19 and 22)
80mm K series profile for both primary and secondary profiles
For 3 kN/m^2 Grid = 1300mm X 1300mm
For 5 kN/m^2 Grid = 1100mm X 1100mm

2- 80mm R Series Profile: 80mm Aluminium Profile Grade A 6063 T-5

2A. The General Case Scenario: (sketch sheets 25 and 28)
Based on a 3 kN/m^2 using 50mm pavers the span between pedestal supports for the 80mm R Series profile will span 1900mm. Based on 5 kN/m^2 using 50mm pavers the 80mm profile will span 1600mm

2B. The Occasional Case Scenario: (sketch sheets 31 and 34)
80mm R Series primary profile and 35mm secondary profile. checks are based on the live loads and the dead load of Pavers.
Limiting factors for spacing of secondary 35mm profiles are 900mm for 3 kN/m^2 and 600mm for 5 kN/m^2 as advised by Betterdeck.
Max span between pedestal supports for the 80mm profile is 1300mm for 3 kN/m^2 and 1300mm for the 5kN/m^2.

2C. The Additional Case Scenario: Aluminium Profile Design 80mm X 80mm R Series Profile Grid: (sketch sheets 37 and 40)
80mm R series profile for both primary and secondary profiles
For 3 kN/m^2 Grid = 1200mm X 1200mm
For 5 kN/m^2 Grid = 1000mm X 1000mm

3- 35mm R Series Profile: 35mm Aluminium Profile Grade A 6063 T-5

3A. The General Case Scenario: (sketch sheets 43 and 46)
Based on a 3 kN/m^2 using 50mm pavers the span between pedestal supports for the 35mm R Series profile will span 900mm. Based on 5 kN/m^2 using 50mm pavers the 35mm R Series profile will span 600mm

3B. The Occasional Case Scenario: (sketch sheets 49 and 52)
35mm R Series primary profile and 35mm secondary profile. checks are based on the live loads and the dead load of Pavers.
Limiting factors for spacing of secondary 35mm R Series profiles are 900mm for 3 kN/m^2 and 600mm for 5 kN/m^2 as advised by Betterdeck.
Max span between pedestal supports for the 35mm profile is 700mm for 3 kN/m^2 and 800mm for the 5kN/m^2.
Note: Since The occasional Case Scenario has the 35mm profile for primary and secondary joists. Therefore the additional case scenario would be the same as the occasional case scenario.

4- 35mm K Series Profile: 35mm Aluminium Profile K Series Grade A 6063 T-5

4A. The General Case Scenario: (sketch sheets 55 and 58)
Based on a 3 kN/m^2 using Composite Decking, the span between pedestal supports for the 35mm K Series profile will span 900mm. Based on 5 kN/m^2 using Composite Decking the 35mm K Series profile will span 600mm

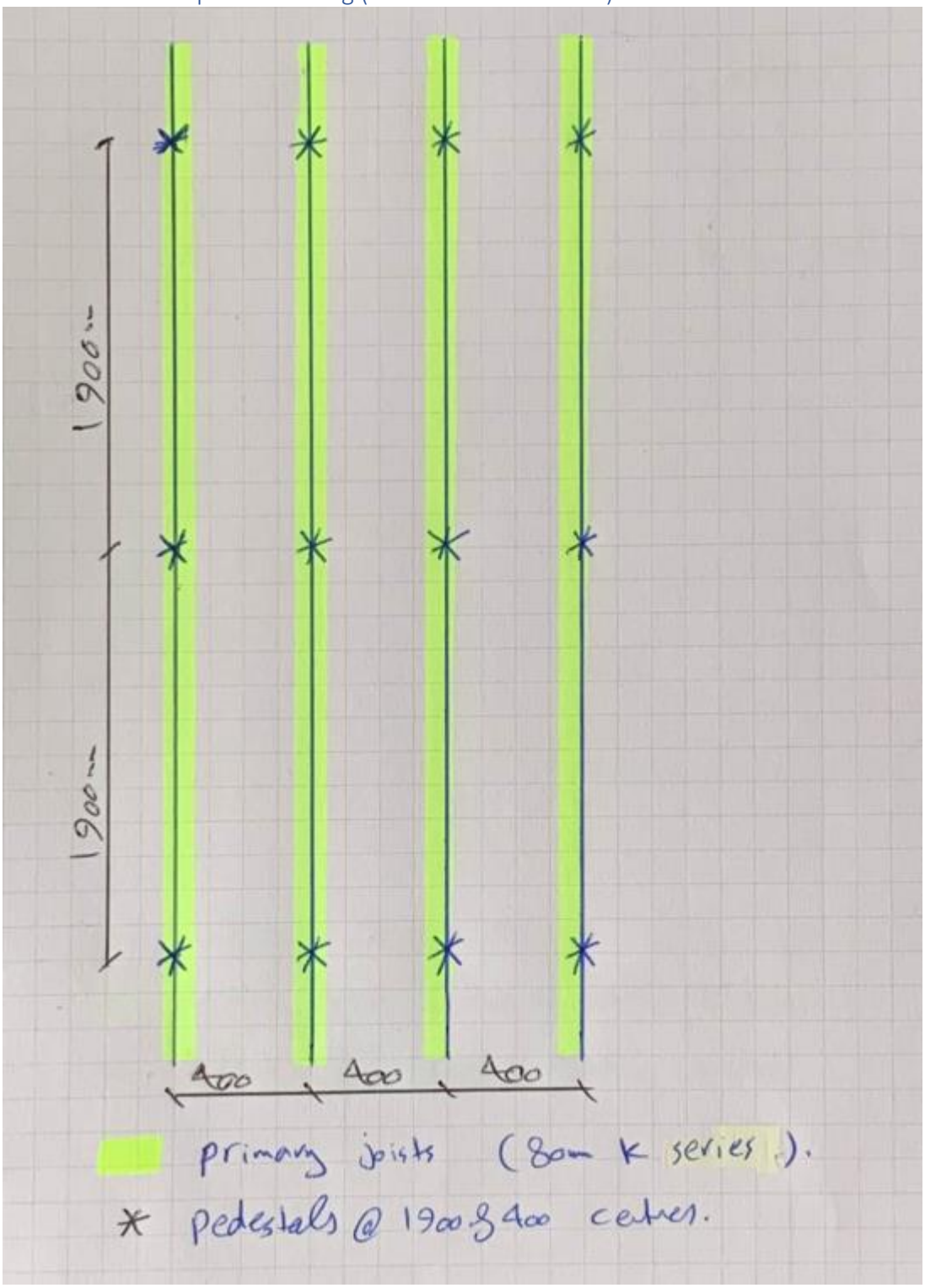
4B. The Occasional Case Scenario: (sketch sheets 61 and 64)
35mm K Series primary profile and 35mm secondary profile. checks are based on the live loads and the dead load of Composite Decking.
Limiting factors for spacing of secondary 35mm profiles are 900mm for 3 kN/m^2 and 600mm for 5 kN/m^2 as advised by Betterdeck.
Max span between pedestal supports for the 35mm profile is 700mm for 3 kN/m^2 and 800mm for the 5kN/m^2.
Note: Since The occasional Case Scenario has the 35mm profile for primary and secondary joists. Therefore the additional case scenario would be the same as the occasional case scenario.

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Project: Aluminium Profile	Contract: 1513-1
Subject: Aluminium Profile Design	Sheet No. 7
Date: 09/04/2021	By: A.N

Aluminium Profile Design For the General Case Scenario (80mm Profile K Series):

Sketch Of Grid Line System For the General Case Scenario For 3kN/m2 Live load and Dead load of Composite Decking (80mm Profile K Series):



Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 8
Date: 09/04/2021	**By:** A.N

Aluminium Profile Checked for Live load of 3kN/m2 and Composite Deck for the General Case Scenario (80mm Profile K Series):

The Profile has the following particulars:

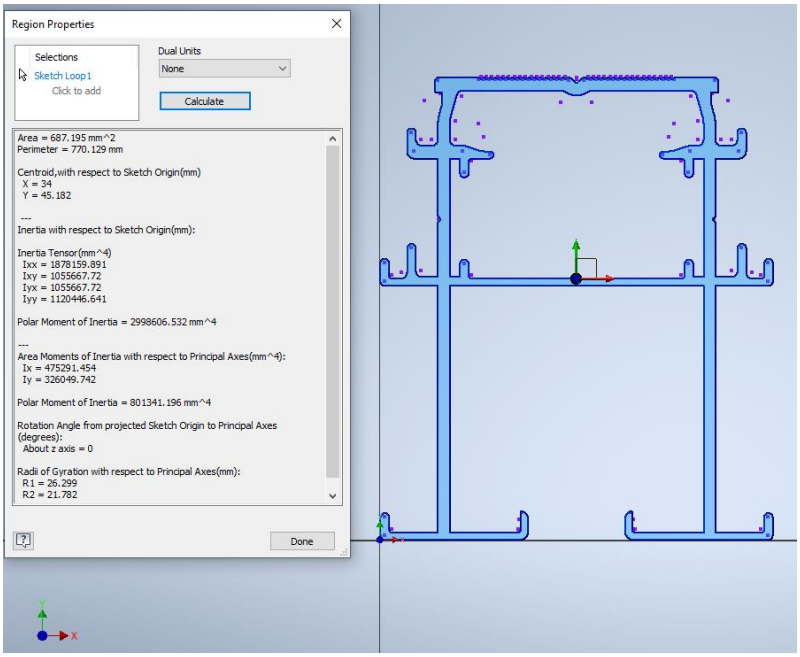
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 475291mm^4$ (80mm K series Profile)

$Z = 10520mm^3$ (80mm K series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



80mm Profiles at Every 400mm C/C.

80mm Profile Span is 1900mm (Profile Length)

Live Load:

UDL Live load $= 3\text{kN/m}^2 \times 0.4\text{m} = 1.2\text{kN/m}$ (SLS)

Dead Load for Composite Deck:

Dead load due to Composite Deck $\rightarrow$ Convert $\frac{\text{lb}}{\text{m}}$ to $\frac{\text{lb}}{\text{m}^2} \rightarrow \frac{7.3\text{lb/m}}{0.136\text{m}} = 55.15\text{lb/m}^2$

Dead load due to Composite Deck $\rightarrow$ Convert $\frac{\text{lb}}{\text{m}^2}$ to $\frac{\text{kN}}{\text{m}^2} \rightarrow 55.15\text{lb/m}^2 = 0.245\text{kN/m}^2$

UDL Dead load due to Composite Deck: $0.245\text{kN/m}^2 \times 0.4\text{m} = 0.098\text{kN/m}$ (SLS)

Total UDL Load: $(1.2\text{kN/m} \times 1.5) + (0.098\text{kN/m} \times 1.35) = 1.9323\text{kN/m}$ (ULS)

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 9
Date: 09/04/2021	**By:** A.N

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{1.9323 \times 1.9^2}{8} = 0.87\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{0.87 \times 10^6}{10520} = 83 \frac{N}{mm^2} < 130 \frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

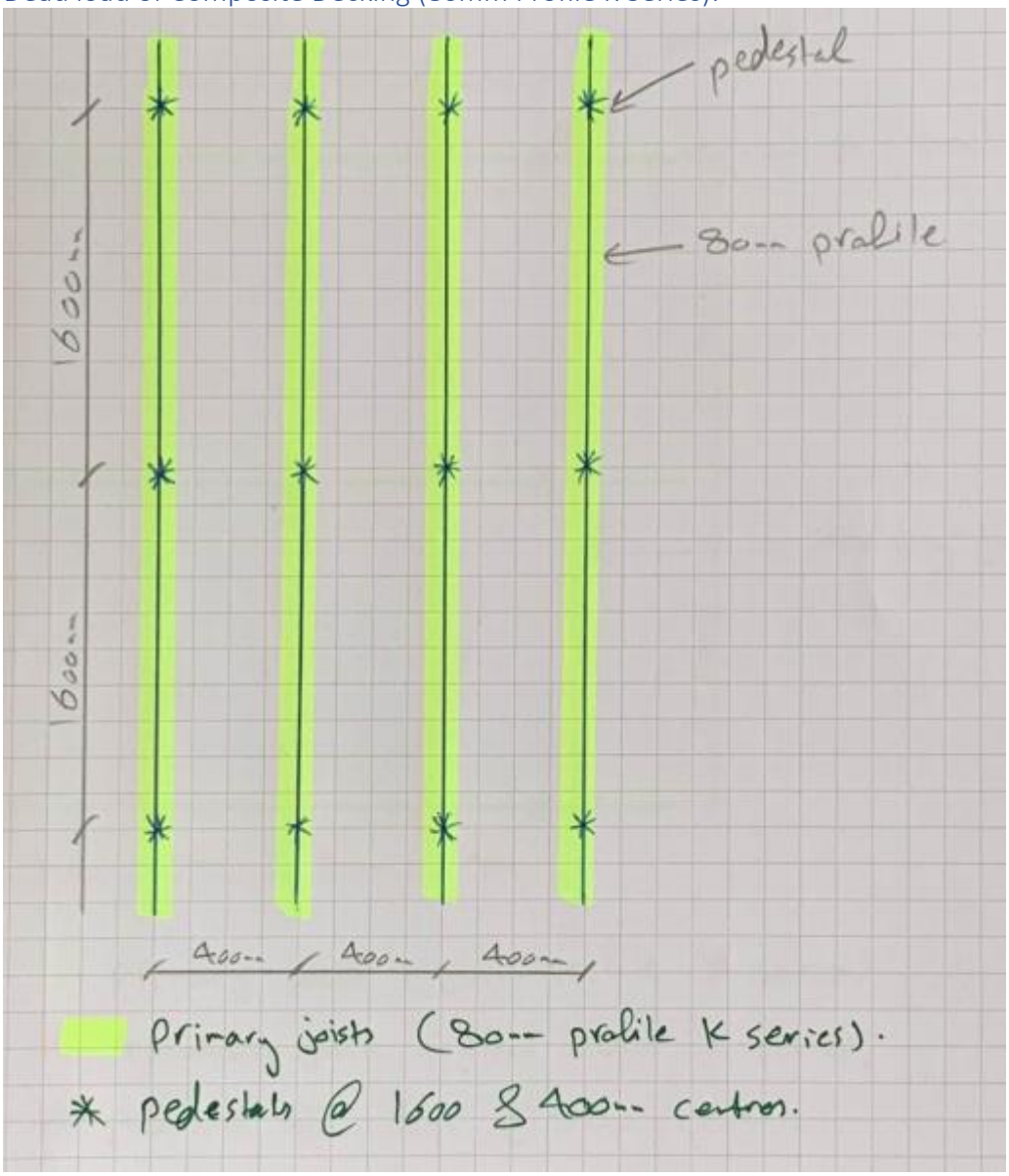
$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 1.298 \times 1900^4}{384 \times 70000 \times 475291} = 6.62\text{mm} < 9.5\text{mm} \rightarrow \left(\frac{1900}{200}\right)$$ **Okay**

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 10
Date: 09/04/2021	**By:** A.N

Sketch Of Grid Line System For the General Case Scenario For 5kN/m2 Live load and Dead load of Composite Decking (80mm Profile K Series):



Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Composite Decking for the General Case Scenario (80mm Profile K Series):

The Profile has the following particulars:

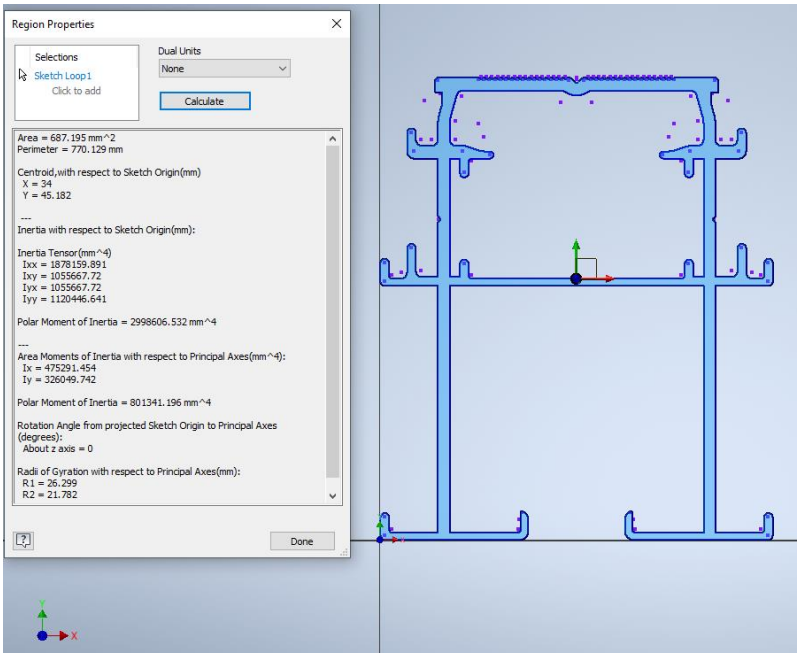
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 475291mm^4$ (80mm K series Profile)

$Z = 10520mm^3$ (80mm K series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



Span Between 80mm Profiles = 400mm

Span Between pedestals = 1600mm (Profile Length)

Live Load:

$\text{UDL Live load} = 5\text{kN/m}^2 \times 0.4\text{m} = 2\text{kN/m}$

Dead Load for Composite Deck:

$\text{Dead load due to Composite Deck} \rightarrow \text{Convert}\frac{\text{lb}}{\text{m}}\text{to}\frac{\text{lb}}{\text{m}^2} \rightarrow \frac{7.3\text{lb/m}}{0.136\text{m}} = 55.15\text{lb/m}^2$

$\text{Dead load due to Composite Deck} \rightarrow \text{Convert}\frac{\text{lb}}{\text{m}^2}\text{to}\frac{\text{kN}}{\text{m}^2} \rightarrow 55.15\text{lb/m}^2 = 0.245\text{kN/m}^2$

$\text{UDL Dead load due to Composite Deck: } 0.245\text{kN/m}^2 \times 0.4\text{m} = 0.098\text{kN/m}$ (SLS)

$\text{Total UDL Load: } (2\text{kN/m} \times 1.5) + (0.098\text{kN/m} \times 1.35) = 3.13\text{kN/m}$ (ULS)

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Project: Aluminium Profile	Contract: 1513-1
Subject: Aluminium Profile Design	Sheet No. 12
Date: 09/04/2021	By: A.N

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{3.13 \times 1.6^2}{8} = 1\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{1 \times 10^6}{10520} = 96 \frac{N}{mm^2} < 130 \frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

$$\alpha_H = \frac{5wl^4}{384EI}$$

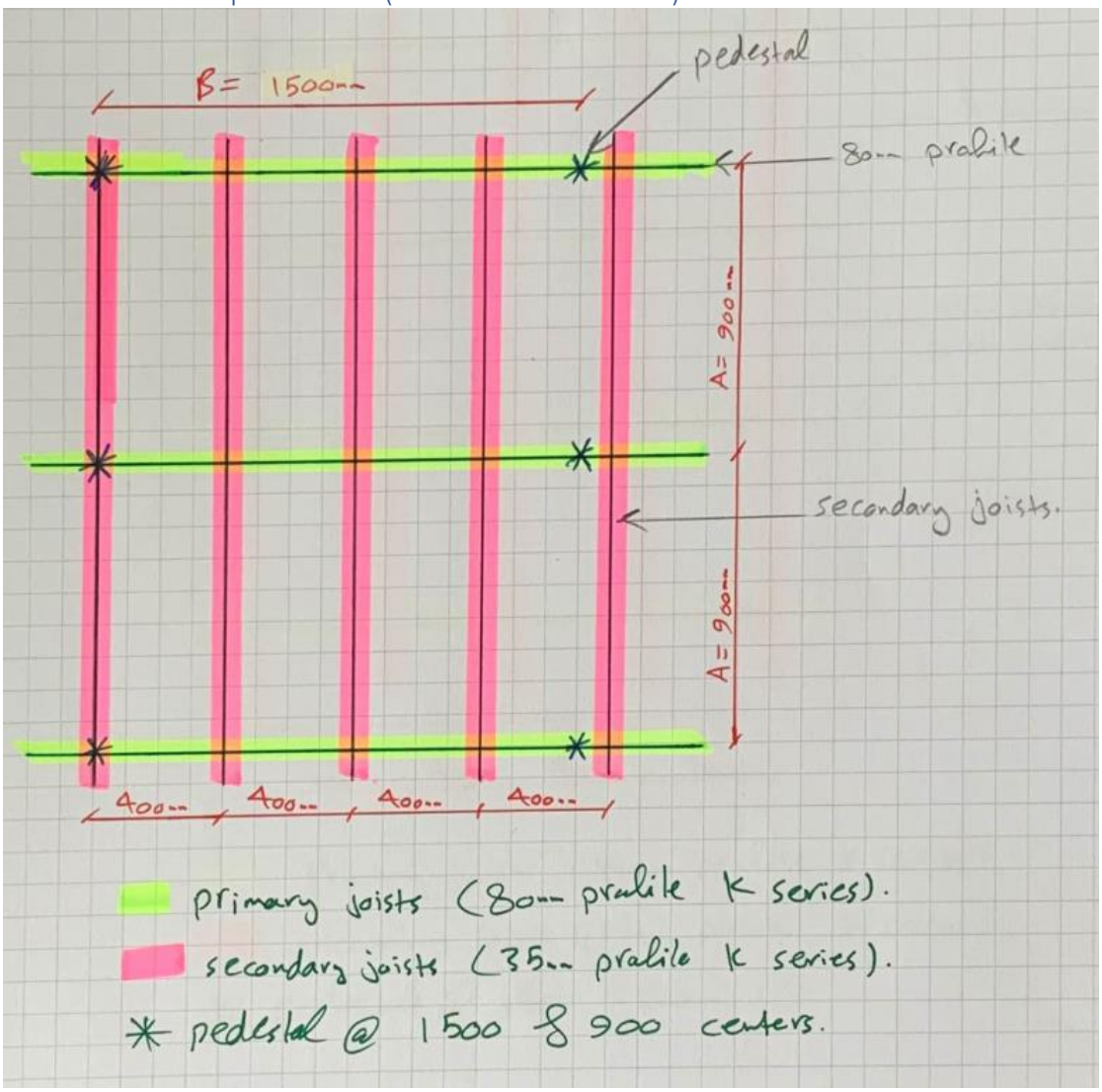
$$\alpha_H = \frac{5 \times 2.098 \times 1600^4}{384 \times 70000 \times 475291} = 5.4\text{mm} < 8\text{mm} \rightarrow \left(\frac{1600}{200}\right)$$ **Okay**

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Project: Aluminium Profile	Contract: 1513-1
Subject: Aluminium Profile Design	Sheet No. 13
Date: 09/04/2021	By: A.N

Aluminium Profile Design For the Occasional Case Scenario (80mm Profile K Series):

Sketch Of Grid Line System For the Occasional Case Scenario for 3kN/m2 Live load and Dead load of Composite Deck (80mm Profile K Series):



Aluminium Profile Checked for Live load of 3kN/m2 and Deal load of the composite Decking for the Occasional Case Scenario (80mm Profile K Series):

The Profile has the following particulars:

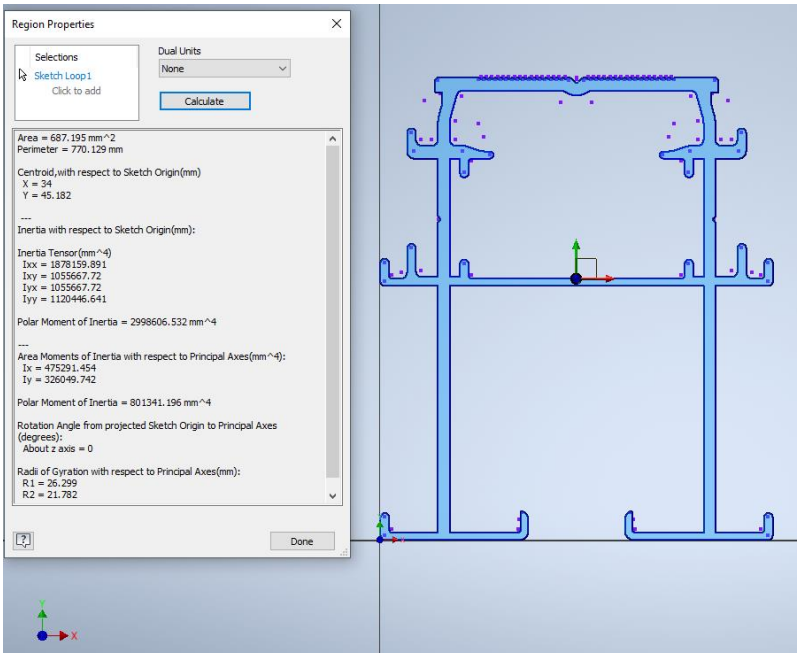
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 475291mm^4$ (80mm K series Profile)

$Z = 10520mm^3$ (80mm K series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



A. Span of secondary joists Assuming (35mm profile) is 900mm (Profile Length)

B. Span of 80mm Profiles is 1500mm (Profile Length)

Live Load:

$\text{UDL Live load} = 3\text{kN/m}^2 \times 0.9\text{m} = 2.7\text{kN/m}\ \text{(SLS)}$

Dead Load for Composite Deck:

$\text{Dead load due to Composite Deck} \rightarrow \text{Convert}\frac{\text{lb}}{\text{m}}\text{to}\frac{\text{lb}}{\text{m}^2} \rightarrow \frac{7.3\text{lb/m}}{0.136\text{m}} = 55.15\text{lb/m}^2$

$\text{Dead load due to Composite Deck} \rightarrow \text{Convert}\frac{\text{lb}}{\text{m}^2}\text{to}\frac{\text{kN}}{\text{m}^2} \rightarrow 55.15\text{lb/m}^2 = 0.245\text{kN/m}^2$

$\text{UDL Dead load due to Composite Deck: } 0.245\text{kN/m}^2 \times 0.9\text{m} = 0.22\text{kN/m}\ \text{(SLS)}$

Dead Load For Secondary Joists spanning 900mm:

Dead load due to secondary Joists: $3\text{Nr} \times 0.000457m^2 \times 27\text{kN/m}^3 = 0.037\text{kN/m}$

Total UDL Load: $(0.22\text{kN/m} + 0.037\text{kN/m}) = 0.257\text{kN/m}$ (ULS)

Total UDL Load: $(2.7\text{kN/m} \times 1.5) + (0.257\text{kN/m} \times 1.35) = 4.397\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{4.397 \times 1.5^2}{8} = 1.23\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{1.23 \times 10^6}{10520} = 117\,\frac{N}{mm^2} < 130\,\frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

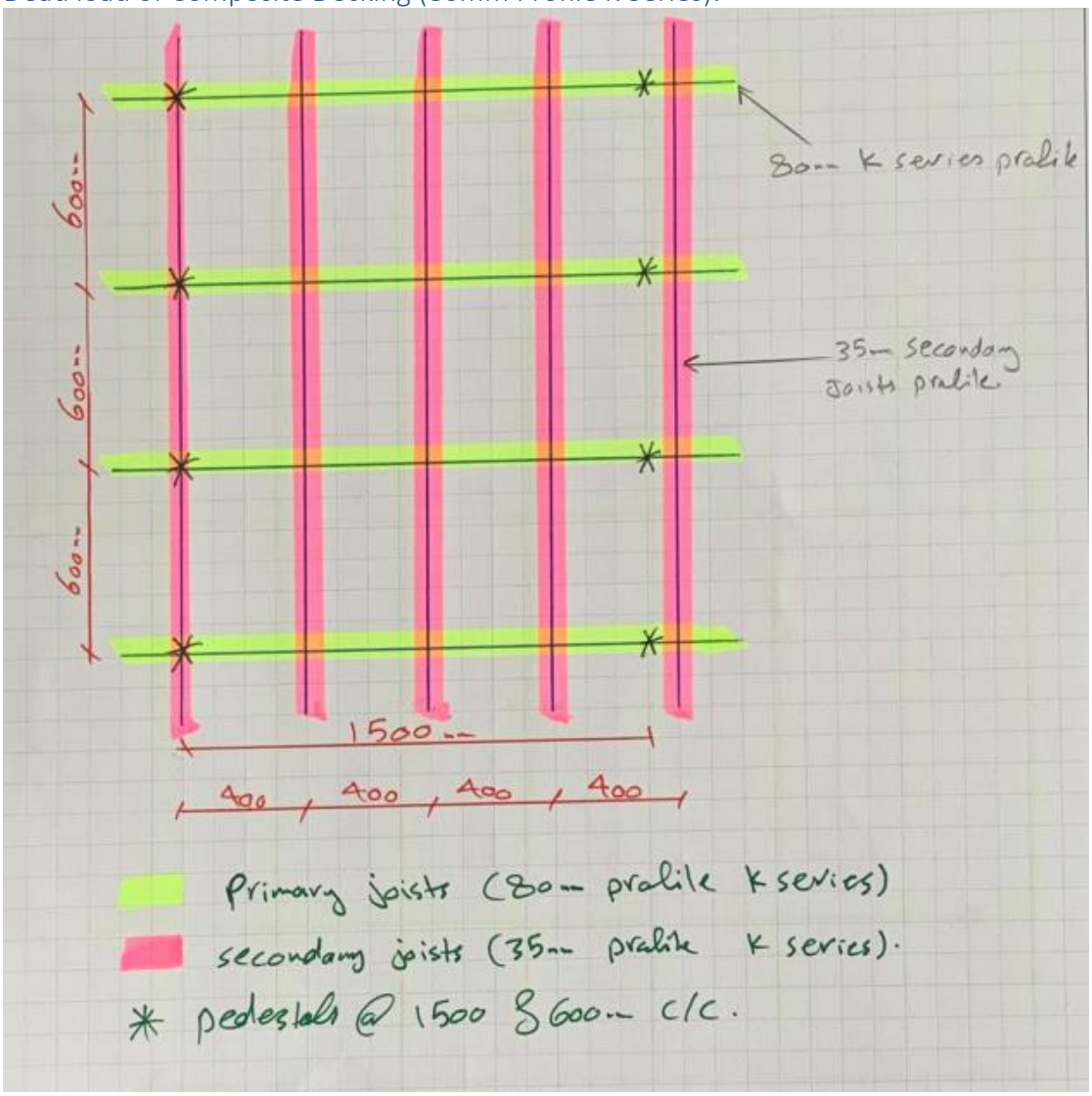
$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 2.957 \times 1500^4}{384 \times 70000 \times 475291} = 5.86\text{mm} < 7.5\text{mm} \rightarrow \left(\frac{1500}{200}\right)$$ **Okay**

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 16
Date: 09/04/2021	**By:** A.N

Sketch Of Grid Line System For the Occasional Case Scenario for 5kN/m2 Live load and Dead load of Composite Decking (80mm Profile K Series):



Aluminium Profile Checked for Live load of 5kN/m2 and Dead Load of the composite Decking for the Occasional Case Scenario (80mm Profile K Series):

The Profile has the following particulars:

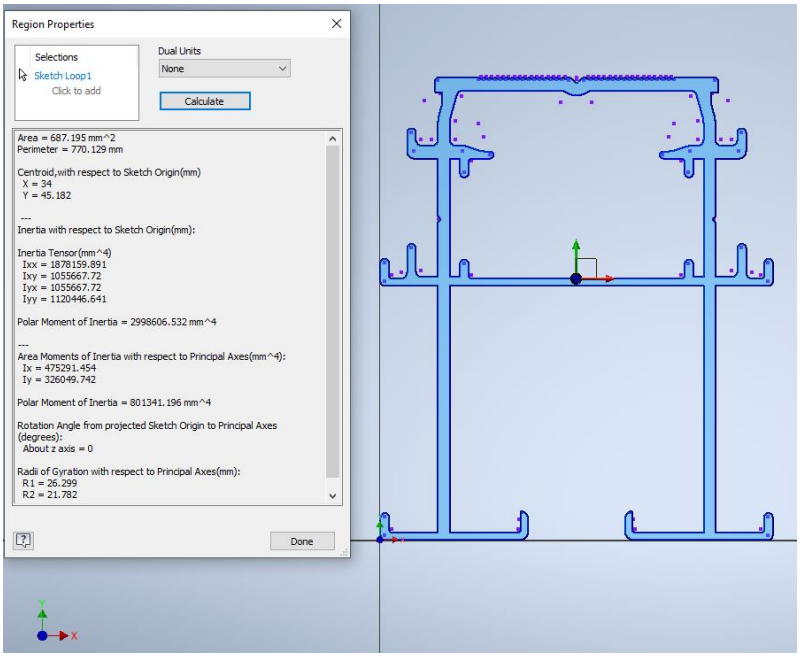
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 475291mm^4$ (80mm K series Profile)

$Z = 10520mm^3$ (80mm K series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



A- Span of secondary joists Assuming (35mm profile) is 600mm (Profile Length)

B- Span of 80mm Profiles is 1500mm (Profile Length)

Live Load:

$$\text{UDL Live load} = 5\text{kN/m}^2 \times 0.6\text{m} = 3\text{kN/m}$$

Dead Load for Composite Deck:

$$\text{Dead load due to Composite Deck} \rightarrow \text{Convert}\frac{\text{lb}}{\text{m}}\text{to}\frac{\text{lb}}{\text{m}^2} \rightarrow \frac{7.3\text{lb/m}}{0.136\text{m}} = 55.15\text{lb/m}^2$$

$$\text{Dead load due to Composite Deck} \rightarrow \text{Convert}\frac{\text{lb}}{\text{m}^2}\text{to}\frac{\text{kN}}{\text{m}^2} \rightarrow 55.15\text{lb/m}^2 = 0.245\text{kN/m}^2$$

$$\text{UDL Dead load due to Composite Deck: } 0.245\text{kN/m}^2 \times 0.6\text{m} = 0.147\text{kN/m} \text{ (SLS)}$$

Dead Load For Secondary Joists spanning 600mm:

Dead load due to secondary Joists: $2\text{Nr} \times 0.000457m^2 \times 27\text{kN/m}^3 = 0.024\text{kN/m}$

Total UDL Dead Load: $0.147\text{kN/m} + 0.024\text{kN/m} = 0.171\text{kN/m}$ (SLS)

Total UDL Load: $(3\text{kN/m} \times 1.5) + (0.171\text{kN/m} \times 1.35) = 4.73\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{4.73 \times 1.5^2}{8} = 1.33\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{0.89 \times 10^6}{10520} = 85\frac{N}{mm^2} < 130\frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

$$\alpha_H = \frac{5wl^4}{384EI}$$

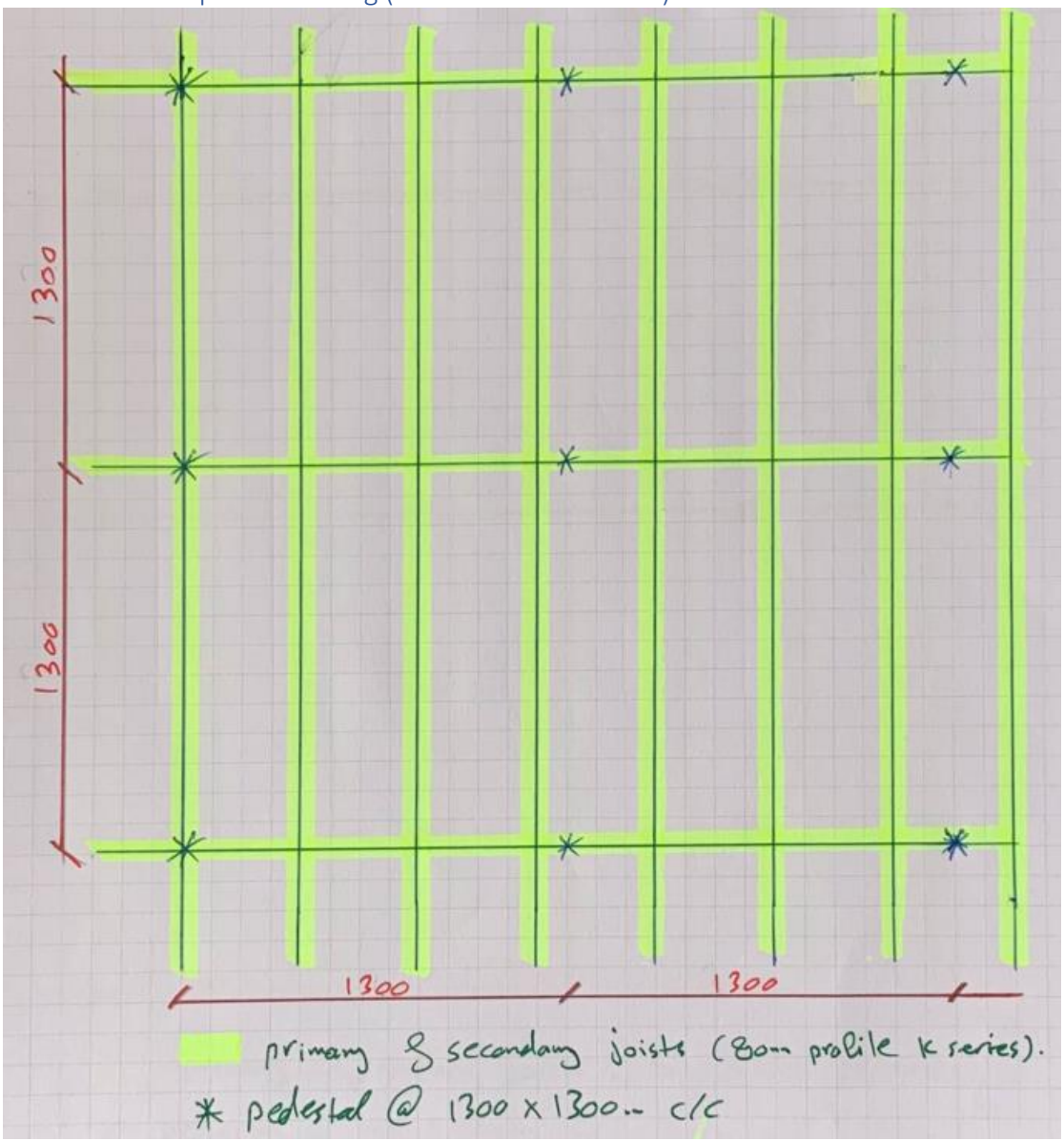
$$\alpha_H = \frac{5 \times 3.171 \times 1500^4}{384 \times 70000 \times 475291} = 6.3\text{mm} < 7.5\text{mm} \rightarrow \left(\frac{1500}{200}\right)$$ **Okay**

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 19
Date: 09/04/2021	**By:** A.N

Aluminium Profile Design 80mm X 80mm Profile Grid (80mm Profile K Series):

Sketch Of Grid Line System For the Last Case Scenario for 3kN/m2 Live load and Dead load of the Composite Decking (80mm Profile K Series):



Aluminium Profile Checked for Live load of 3kN/m2 and Composite Deck for the Last Case Scenario (80mm Profile K Series):

The Profile has the following particulars:

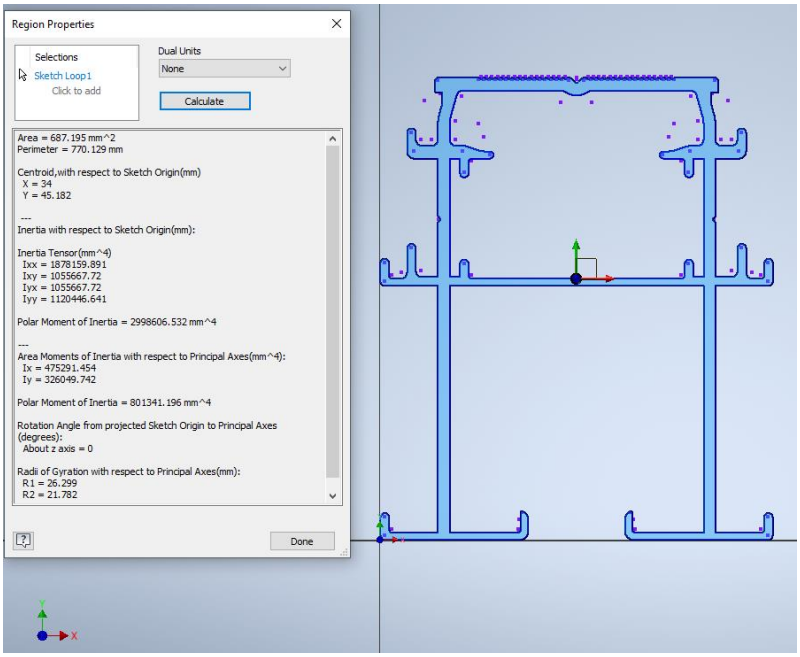
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 475291mm^4$ (80mm K series Profile)

$Z = 10520mm^3$ (80mm K series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



A- 80mm Profiles at Every 1300mm C/C.

B- Span between pedestals is 1300mm C/C (Profile Length)

Live Load:

$\text{UDL Live load} = 3\text{kN/m}^2 \times 1.3\text{m} = 3.9\text{kN/m}\ \text{(SLS)}$

Dead Load for Composite Deck:

$\text{Dead load due to Composite Deck} \rightarrow \text{Convert}\frac{\text{lb}}{\text{m}}\text{to}\frac{\text{lb}}{\text{m}^2} \rightarrow \frac{7.3\text{lb/m}}{0.136\text{m}} = 55.15\text{lb/m}^2$

$\text{Dead load due to Composite Deck} \rightarrow \text{Convert}\frac{\text{lb}}{\text{m}^2}\text{to}\frac{\text{kN}}{\text{m}^2} \rightarrow 55.15\text{lb/m}^2 = 0.245\text{kN/m}^2$

$\text{UDL Dead load due to Composite Deck: } 0.245\text{kN/m}^2 \times 1.3\text{m} = 0.32\text{kN/m}\ \text{(SLS)}$

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Project: Aluminium Profile	Contract: 1513-1
Subject: Aluminium Profile Design	Sheet No. 21
Date: 09/04/2021	By: A.N

Dead Load For 80mm Profile spanning 1300mm:

Dead load due to secondary Joists: $2\text{Nr} \times (0.000722\text{m}^2) \times 27\text{kN/m}^3 = 0.039\text{kN}$

Total UDL Load: $0.32 + 0.039 = 0.359\text{kN/m}$ (SLS)

Total UDL Load: $(3.9\text{kN/m} \times 1.5) + (0.359\text{kN/m} \times 1.35) = 6.27\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{6.27 \times 1.3^2}{8} = 1.325\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{1.325 \times 10^6}{10520} = 126\,\frac{N}{mm^2} < 130\,\frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

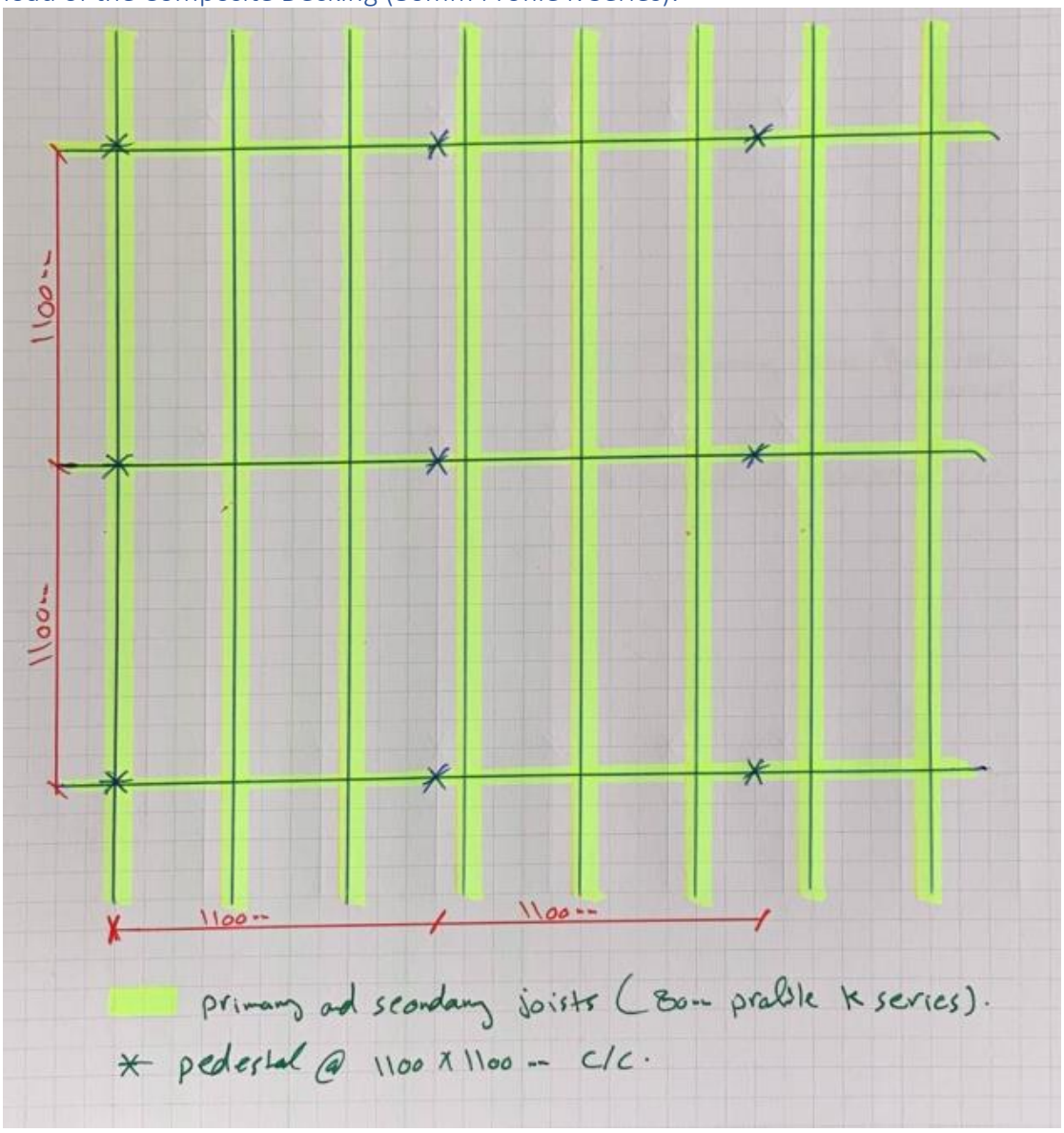
$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 4.259 \times 1300^4}{384 \times 70000 \times 475291} = 4.8\text{mm} < 6.5\text{mm} \rightarrow \left(\frac{1300}{200}\right)$$ **Okay**

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 22
Date: 09/04/2021	**By:** A.N

Sketch Of Grid Line System For the Last Case Scenario for 5kN/m2 Live load and Dead load of the Composite Decking (80mm Profile K Series):



Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Composite Decking for the Last Case Scenario (80mm Profile K Series):

The Profile has the following particulars:

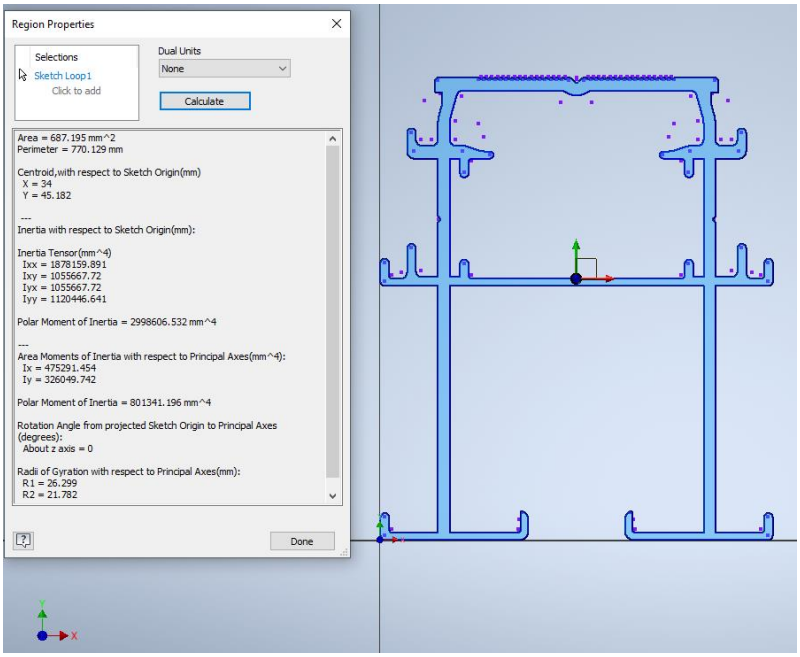
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 475291mm^4$ (80mm K series Profile)

$Z = 10520mm^3$ (80mm K series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



A- 80mm Profiles at Every 1100mm C/C.

B- Span between pedestals is 1100mm C/C (Profile Length)

Live Load:

$$\text{UDL Live load} = 5\text{kN/m}^2 \times 1.1\text{m} = 5.5\text{kN/m}$$

Dead Load for Composite Deck:

$$\text{Dead load due to Composite Deck} \rightarrow \text{Convert}\frac{\text{lb}}{\text{m}}\text{to}\frac{\text{lb}}{\text{m}^2} \rightarrow \frac{7.3\text{lb/m}}{0.136\text{m}} = 55.15\text{lb/m}^2$$

$$\text{Dead load due to Composite Deck} \rightarrow \text{Convert}\frac{\text{lb}}{\text{m}^2}\text{to}\frac{\text{kN}}{\text{m}^2} \rightarrow 55.15\text{lb/m}^2 = 0.245\text{kN/m}^2$$

$$\text{UDL Dead load due to Composite Deck: } 0.245\text{kN/m}^2 \times 1.1\text{m} = 0.27\text{kN/m} \text{ (SLS)}$$

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Project: Aluminium Profile	Contract: 1513-1
Subject: Aluminium Profile Design	Sheet No. 24
Date: 09/04/2021	By: A.N

Dead Load For 80mm Profile spanning 1100mm:

Dead load due to secondary Joists: $2\text{Nr} \times (0.000722\text{m}^2) \times 27\text{kN/m}^3 = 0.039\text{kN}$

Total UDL Load: $0.27 + 0.039 = 0.309\text{kN/m}$ (SLS)

Total UDL Load: $(5.5\text{kN/m} \times 1.5) + (0.309\text{kN/m} \times 1.35) = 8.67\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{8.67 \times 1.1^2}{8} = 1.31\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{1.31 \times 10^6}{10520} = 125\,\frac{N}{mm^2} < 130\,\frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

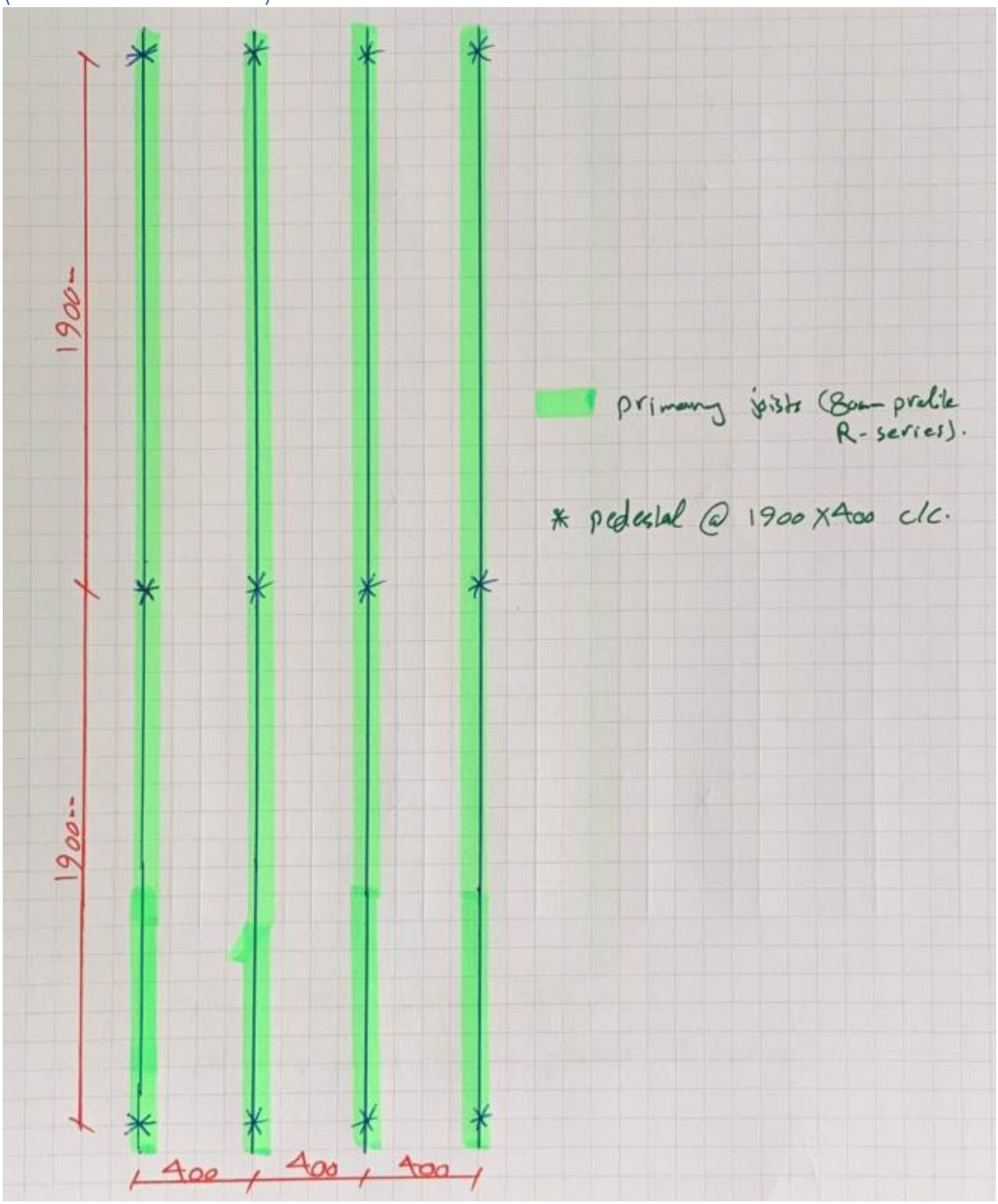
$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 5.809 \times 1100^4}{384 \times 70000 \times 475291} = 3.33\text{mm} < 5.5\text{mm} \rightarrow \left(\frac{1100}{200}\right)$$ **Okay**

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 25
Date: 09/04/2021	**By:** A.N

Aluminium Profile Design For the General Case Scenario (80mm Profile R Series):

Sketch Of Grid Line System For the General Case Scenario For 3kN/m2 Live load (80mm Profile R Series):



Aluminium Profile Checked for Live load of 3kN/m2 and Dead Load of Pavers for the General Case Scenario (80mm Profile R Series):

The Profile has the following particulars:

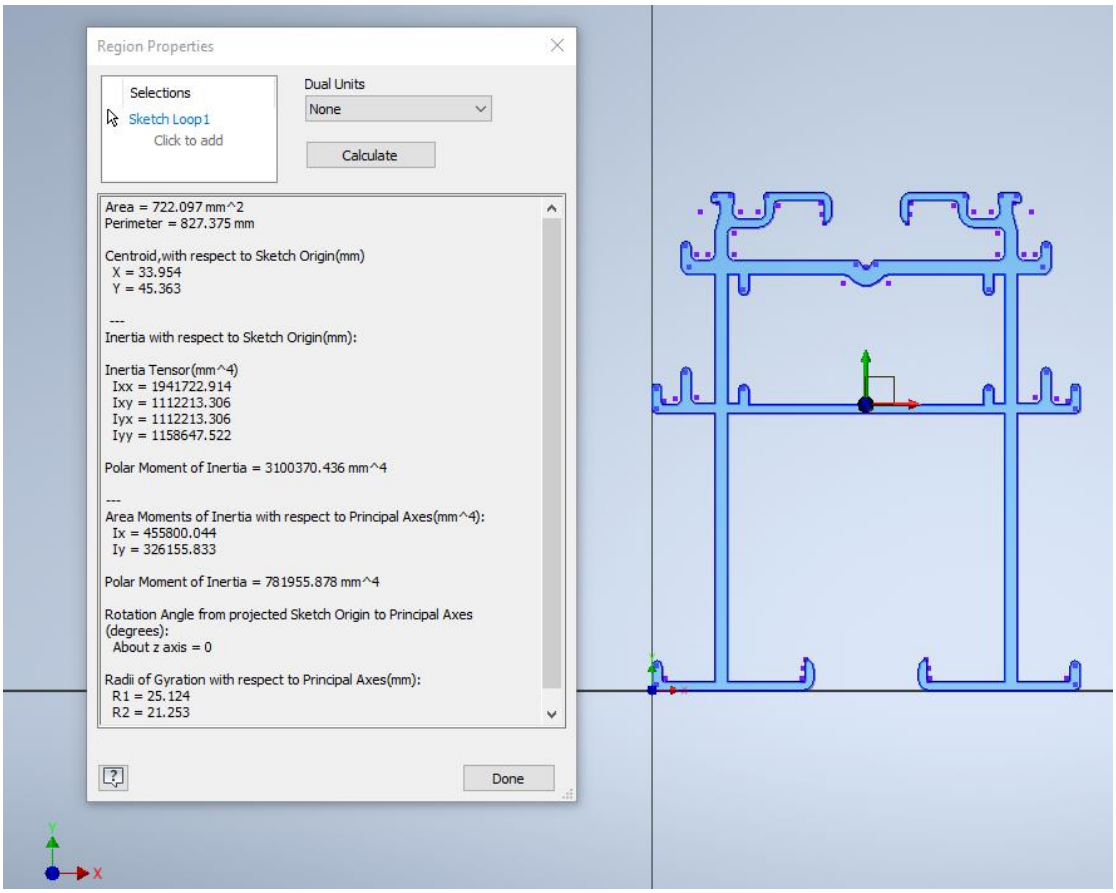
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 455800mm^4$ (80mm R Series Profile)

$Z = 10048mm^3$ (80mm R Series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



80mm Profiles at Every 400mm C/C.

80mm Profile Span is 1900mm (Profile Length)

Live Load:

UDL Live load $= 3\text{kN/m}^2 \times 0.4\text{m} = 1.2\text{kN/m}$ (SLS)

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 27
Date: 09/04/2021	**By:** A.N

Dead Load for Pavers:

Dimensions (mm)	Colours	Pcs per m2	m2 per bale	Pcs per Bale	No. Layers	Approx. Bale Weight (KG)	Profile	Strength
400 x 400 x 40	Silver Granite, Black Granite, Buff Granite	6.25	13.44	84	14	1215	Chamfer	Class 1
450 x 450 x 40	Silver Granite, Black Granite, Buff Granite,	4.94	12.96	64	16	1150	Square edge	Class 1
400 x 400 x 50	Silver Granite, Black Granite, Buff Granite	6.25	11.52	72	12	1215	Chamfer	Class 1

Dead load due to Pavers: Convert from kg to kN $\rightarrow 1215\text{kg} = 11.92\text{kN}$

Dead load due to Pavers: if bale area 11.52m^2 gave us bale weight of 11.92kN,
Therefore $\rightarrow \left(\frac{11.92\text{kN}}{11.52\text{m}^2}\right) = 1.03\text{kN/m}^2$

UDL Dead load due to Pavers: $1.03\text{kN/m}^2 \times 0.4\text{m} = 0.412\text{kN/m}$

Total UDL Load: $(1.2\text{kN/m} \times 1.5) + (0.412\text{kN/m} \times 1.35) = 2.356\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{2.356 \times 1.9^2}{8} = 1.063\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{1.063 \times 10^6}{10048} = 106\,\frac{N}{mm^2} < 130\,\frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

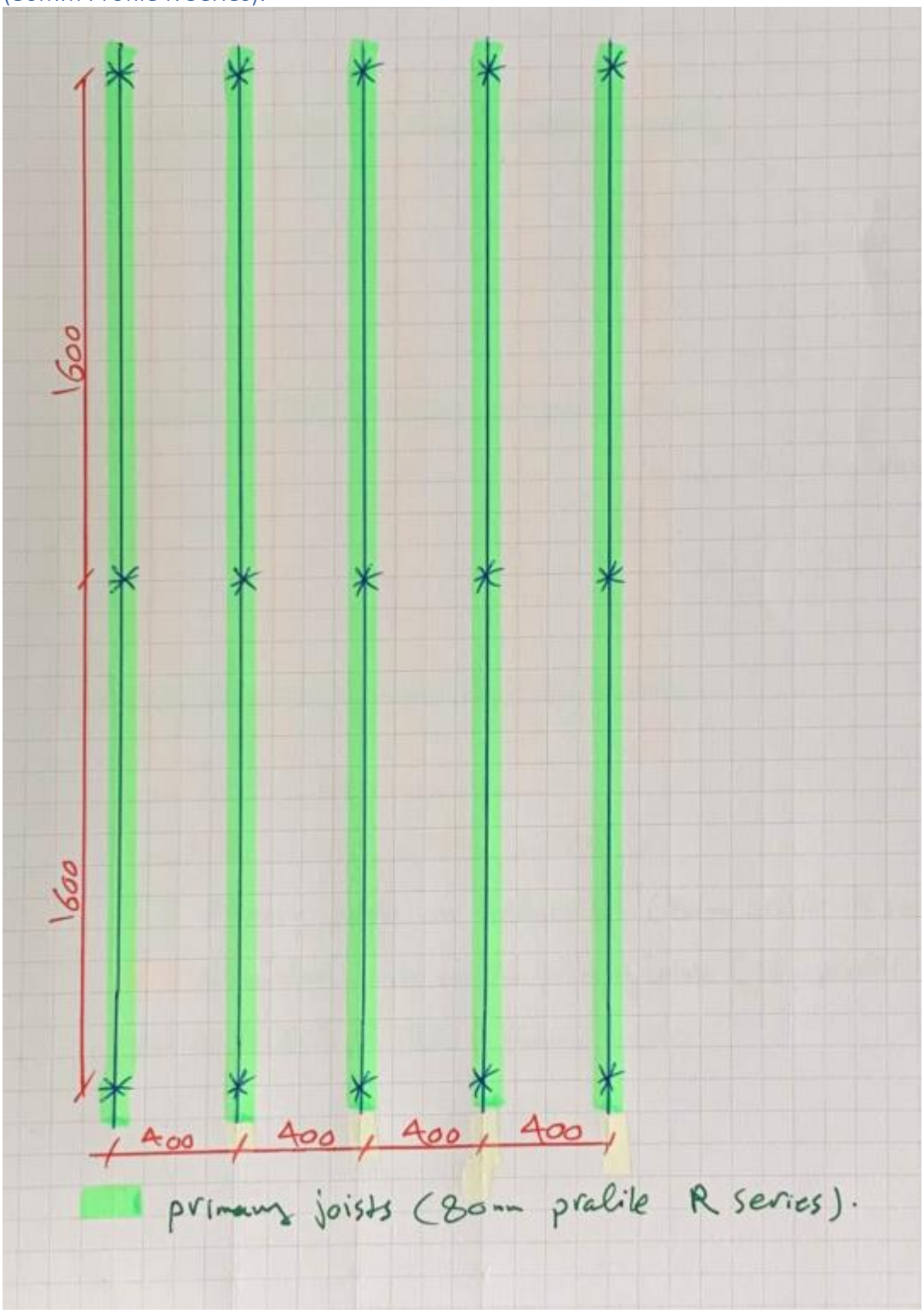
$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 1.612 \times 1900^4}{384 \times 70000 \times 455800} = 8.6\text{mm} < 9.5\text{mm} \rightarrow \left(\frac{1900}{200}\right)$$ **Okay**

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 28
Date: 09/04/2021	**By:** A.N

Sketch Of Grid Line System For the General Case Scenario For 5kN/m2 Live load (80mm Profile R Series):



Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Pavers for the General Case Scenario (80mm Profile R Series):

The Profile has the following particulars:

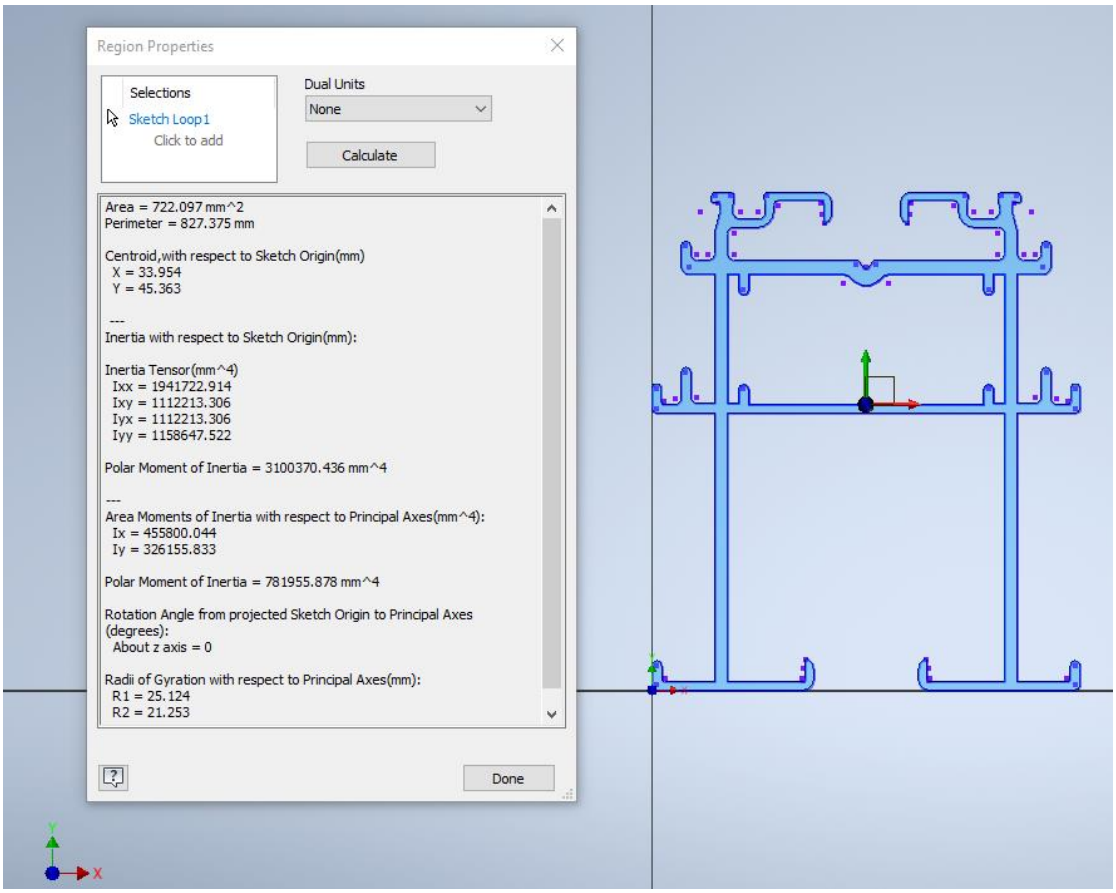
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 455800mm^4$ (80mm R Series Profile)

$Z = 10048mm^3$ (80mm R Series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



Span Between 80mm Profiles = 400mm

Span Between pedestals = 1600mm (Profile Length)

Live Load:

$$\text{UDL Live load} = 5\text{kN/m}^2 \times 0.4\text{m} = 2\text{kN/m}$$

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 30
Date: 09/04/2021	**By:** A.N

Dead Load for Pavers:

Dimensions (mm)	Colours	Pcs per m2	m2 per bale	Pcs per Bale	No. Layers	Approx. Bale Weight (KG)	Profile	Strength
400 x 400 x 40	Silver Granite, Black Granite, Buff Granite	6.25	13.44	84	14	1215	Chamfer	Class 1
450 x 450 x 40	Silver Granite, Black Granite, Buff Granite,	4.94	12.96	64	16	1150	Square edge	Class 1
400 x 400 x 50	Silver Granite, Black Granite, Buff Granite	6.25	11.52	72	12	1215	Chamfer	Class 1

Dead load due to Pavers: Convert from kg to kN $\rightarrow 1215\text{kg} = 11.92\text{kN}$

Dead load due to Pavers: if bale area 11.52m^2 gave us bale weight of 11.92kN,
Therefore $\rightarrow \left(\frac{11.92\text{kN}}{11.52\text{m}^2}\right) = 1.03\text{kN/m}^2$

UDL Dead load due to Pavers: $1.03\text{kN/m}^2 \times 0.4\text{m} = 0.412\text{kN/m}$

Total UDL Load: $(2\text{kN/m} \times 1.5) + (0.412\text{kN/m} \times 1.35) = 3.56\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{3.56 \times 1.6^2}{8} = 1.14\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{1.14 \times 10^6}{10048} = 114\,\frac{N}{mm^2} < 130\,\frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

$$\alpha_H = \frac{5wl^4}{384EI}$$

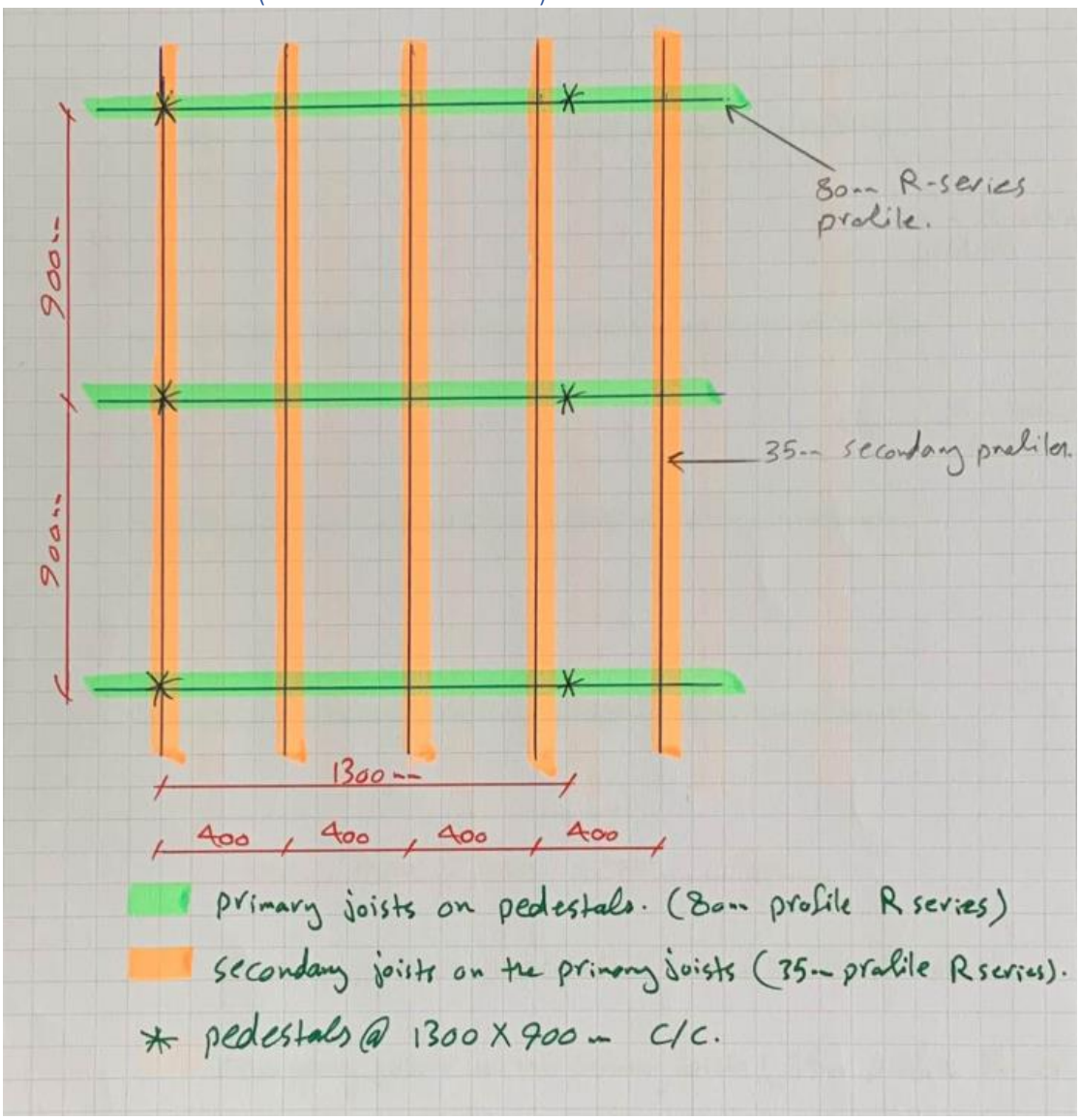
$$\alpha_H = \frac{5 \times 2.412 \times 1600^4}{384 \times 70000 \times 455800} = 6.5\text{mm} < 8\text{mm} \rightarrow \left(\frac{1600}{200}\right)$$ **Okay**

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Project: Aluminium Profile	Contract: 1513-1
Subject: Aluminium Profile Design	Sheet No. 31
Date: 09/04/2021	By: A.N

Aluminium Profile Design For the Occasional Case Scenario (80mm Profile R Series):

Sketch Of Grid Line System For the Occasional Case Scenario for 3kN/m2 Live load and Dead load of Pavers (80mm Profile R Series):



Aluminium Profile Checked for Live load of 3kN/m2 and Dead load of the Pavers the Occasional Case Scenario (80mm Profile R Series):

The Profile has the following particulars:

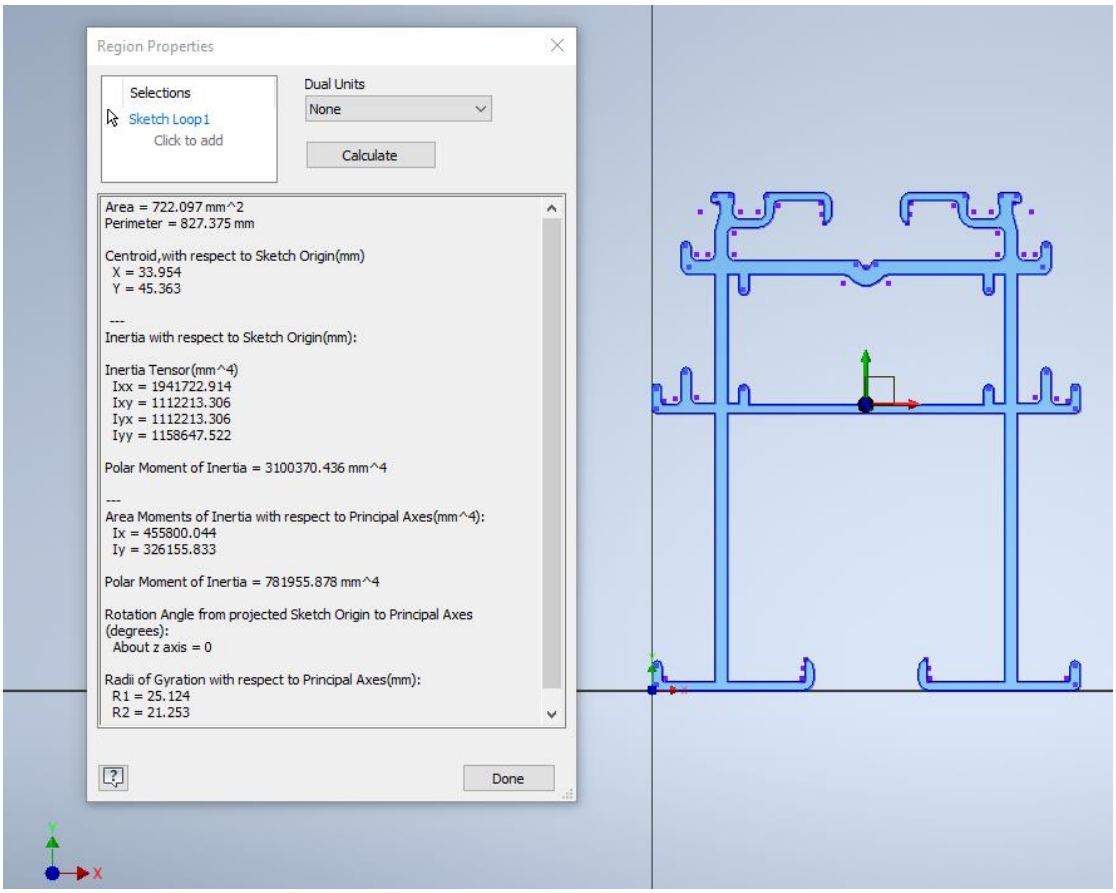
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 455800mm^4$ (80mm R Series Profile)

$Z = 10048mm^3$ (80mm R Series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



Span of secondary joists Assuming (35mm profile) is 900mm (Profile Length)

Span of 80mm Profiles is 1300mm (Profile Length)

Live Load:

$\text{UDL Live load} = 3\text{kN/m}^2 \times 0.9\text{m} = 2.7\text{kN/m}$ (SLS)

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 33
Date: 09/04/2021	**By:** A.N

Dead Load for Pavers:

Dimensions (mm)	Colours	Pcs per m2	m2 per bale	Pcs per Bale	No. Layers	Approx. Bale Weight (KG)	Profile	Strength
400 x 400 x 40	Silver Granite, Black Granite, Buff Granite	6.25	13.44	84	14	1215	Chamfer	Class 1
450 x 450 x 40	Silver Granite, Black Granite, Buff Granite,	4.94	12.96	64	16	1150	Square edge	Class 1
400 x 400 x 50	Silver Granite, Black Granite, Buff Granite	6.25	11.52	72	12	1215	Chamfer	Class 1

Dead load due to Pavers: Convert from kg to kN $\rightarrow 1215\text{kg} = 11.92\text{kN}$

Dead load due to Pavers: if bale area 11.52m^2 gave us bale weight of 11.92kN,
Therefore $\rightarrow \left(\frac{11.92\text{kN}}{11.52\text{m}^2}\right) = 1.03\text{kN/m}^2$

UDL Dead load due to Pavers: $1.03\text{kN/m}^2 \times 0.9\text{m} = 0.927\text{kN/m}$

Dead Load For 35mm Profiles Secondary Joists:

Dead load due to secondary Joists: $3\text{Nr} \times 0.000457m^2 \times 27\text{kN/m}^3 = 0.037\text{kN/m}$

Total UDL Load: $(0.037\text{kN/m} + 0.927\text{kN/m}) = 0.964\text{kN/m}$ (ULS)

Total UDL Load: $(2.7\text{kN/m} \times 1.5) + (0.964\text{kN/m} \times 1.35) = 5.35\text{kN/m}$ (ULS)

Maximum Moment:

$$\text{M}_\text{H} = \frac{\text{w}\times\text{l}^2}{8}$$

$$\text{M}_\text{H} = \frac{5.35\times1.3^2}{8} = 1.13\text{kNm}$$

Maximum Stress:

$$\sigma_{\max} = \frac{\text{M}}{\text{Z}}$$

$$\sigma_{\max} = \frac{1.13\times10^6}{10048} = 113\,\frac{\text{N}}{\text{mm}^2} < 130\,\frac{\text{N}}{\text{mm}^2}$$ **Okay**

Maximum Deflection:

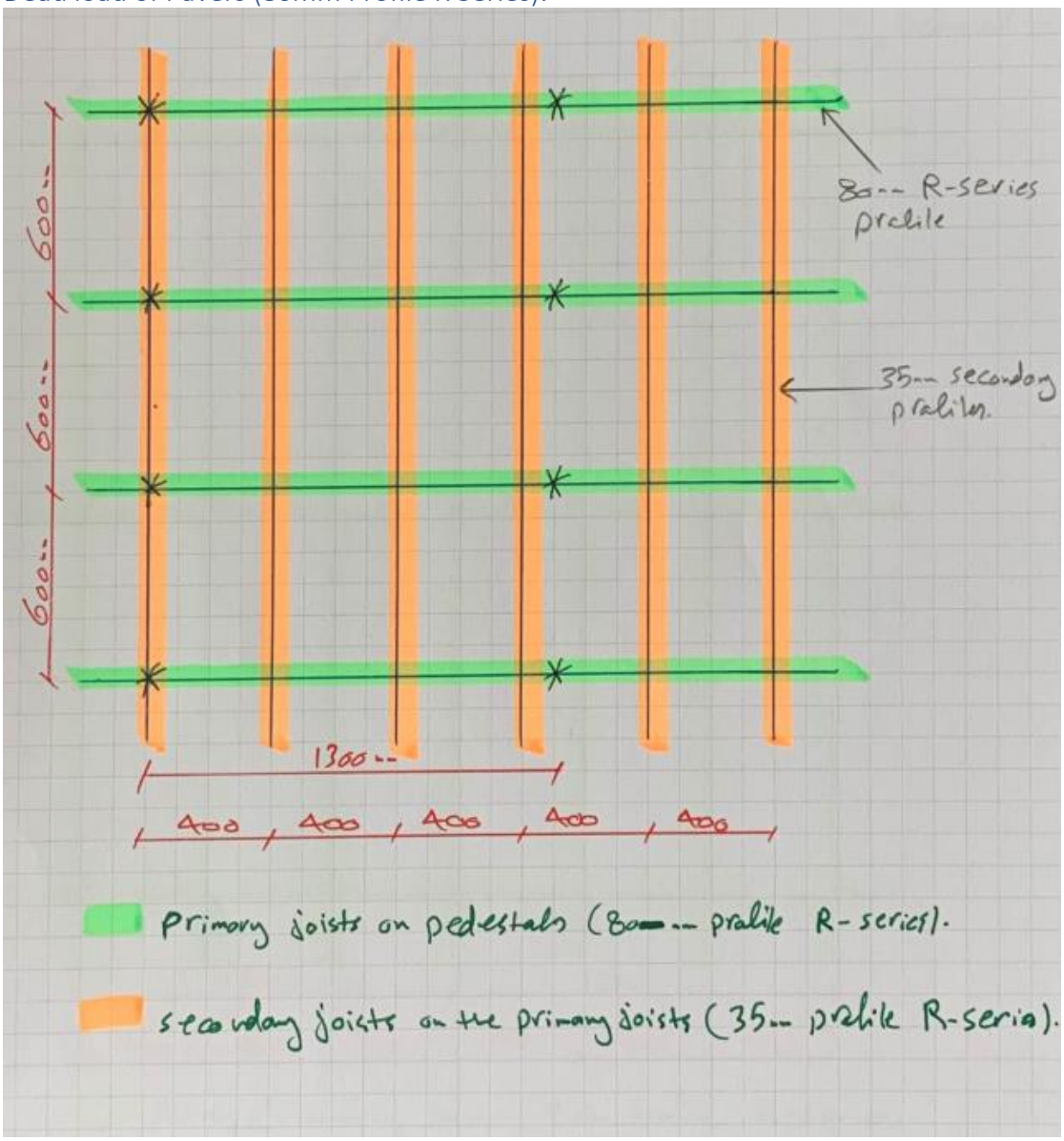
$$\alpha_\text{H} = \frac{5\text{wl}^4}{384\text{EI}}$$

$$\alpha_\text{H} = \frac{5\times3.664\times1300^4}{384\times70000\times455800} = 4.3\text{mm} < 6.5\text{mm} \rightarrow \left(\frac{1300}{200}\right)$$ **Okay**

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 34
Date: 09/04/2021	**By:** A.N

Sketch Of Grid Line System For the Occasional Case Scenario for 5kN/m2 Live load and Dead load of Pavers (80mm Profile R Series):



Aluminium Profile Checked for Live load of 5kN/m2 and Dead Load of the Pavers for the Occasional Case Scenario (80mm Profile R Series):

The Profile has the following particulars:

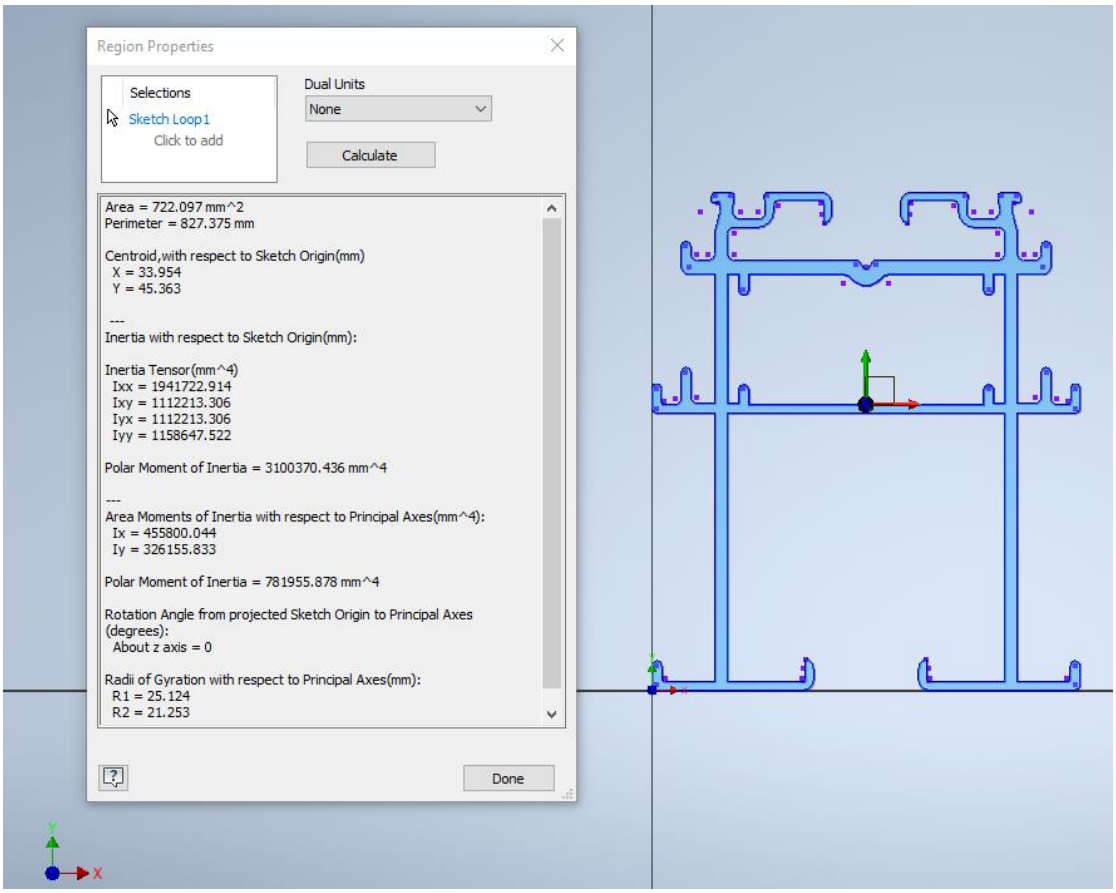
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 455800mm^4$ (80mm R Series Profile)

$Z = 10048mm^3$ (80mm R Series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



Span of secondary joists Assuming (35mm profiles) is 600mm (Profile Length)

Span of 80mm Profiles is 1300mm (Profile Length)

Live Load:

$$\text{UDL Live load} = 5\text{kN/m}^2 \times 0.6\text{m} = 3\text{kN/m}$$

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 36
Date: 09/04/2021	**By:** A.N

Dead Load for Pavers:

Dimensions (mm)	Colours	Pcs per m2	m2 per bale	Pcs per Bale	No. Layers	Approx. Bale Weight (KG)	Profile	Strength
400 x 400 x 40	Silver Granite, Black Granite, Buff Granite	6.25	13.44	84	14	1215	Chamfer	Class 1
450 x 450 x 40	Silver Granite, Black Granite, Buff Granite,	4.94	12.96	64	16	1150	Square edge	Class 1
400 x 400 x 50	Silver Granite, Black Granite, Buff Granite	6.25	11.52	72	12	1215	Chamfer	Class 1

Dead load due to Pavers: Convert from kg to kN $\rightarrow 1215\text{kg} = 11.92\text{kN}$

Dead load due to Pavers: if bale area 11.52m^2 gave us bale weight of 11.92kN,
Therefore $\rightarrow \left(\frac{11.92\text{kN}}{11.52\text{m}^2}\right) = 1.03\text{kN/m}^2$

UDL Dead load due to Pavers: $1.03\text{kN/m}^2 \times 0.6\text{m} = 0.618\text{kN/m}$

Dead Load For 35mm Profiles Secondary Joists:

Dead load due to secondary Joists: $2\text{Nr} \times 0.000457m^2 \times 27\text{kN/m}^3 = 0.024\text{kN/m}$

Total UDL Dead Load: $0.618\text{kN/m} + 0.024\text{kN/m} = 0.642\text{kN/m}$ (SLS)

Total UDL Load: $(3\text{kN/m} \times 1.5) + (0.642\text{kN/m} \times 1.35) = 5.367\text{kN/m}$ (ULS)

Maximum Moment:

$$\text{M}_\text{H} = \frac{\text{w}\times\text{l}^2}{8}$$

$$\text{M}_\text{H} = \frac{5.367\times1.3^2}{8} = 1.133\text{kNm}$$

Maximum Stress:

$$\sigma_{\max} = \frac{\text{M}}{\text{Z}}$$

$$\sigma_{\max} = \frac{1.133\times10^6}{10048} = 113\,\frac{\text{N}}{\text{mm}^2} < 130\,\frac{\text{N}}{\text{mm}^2}$$ **Okay**

Maximum Deflection:

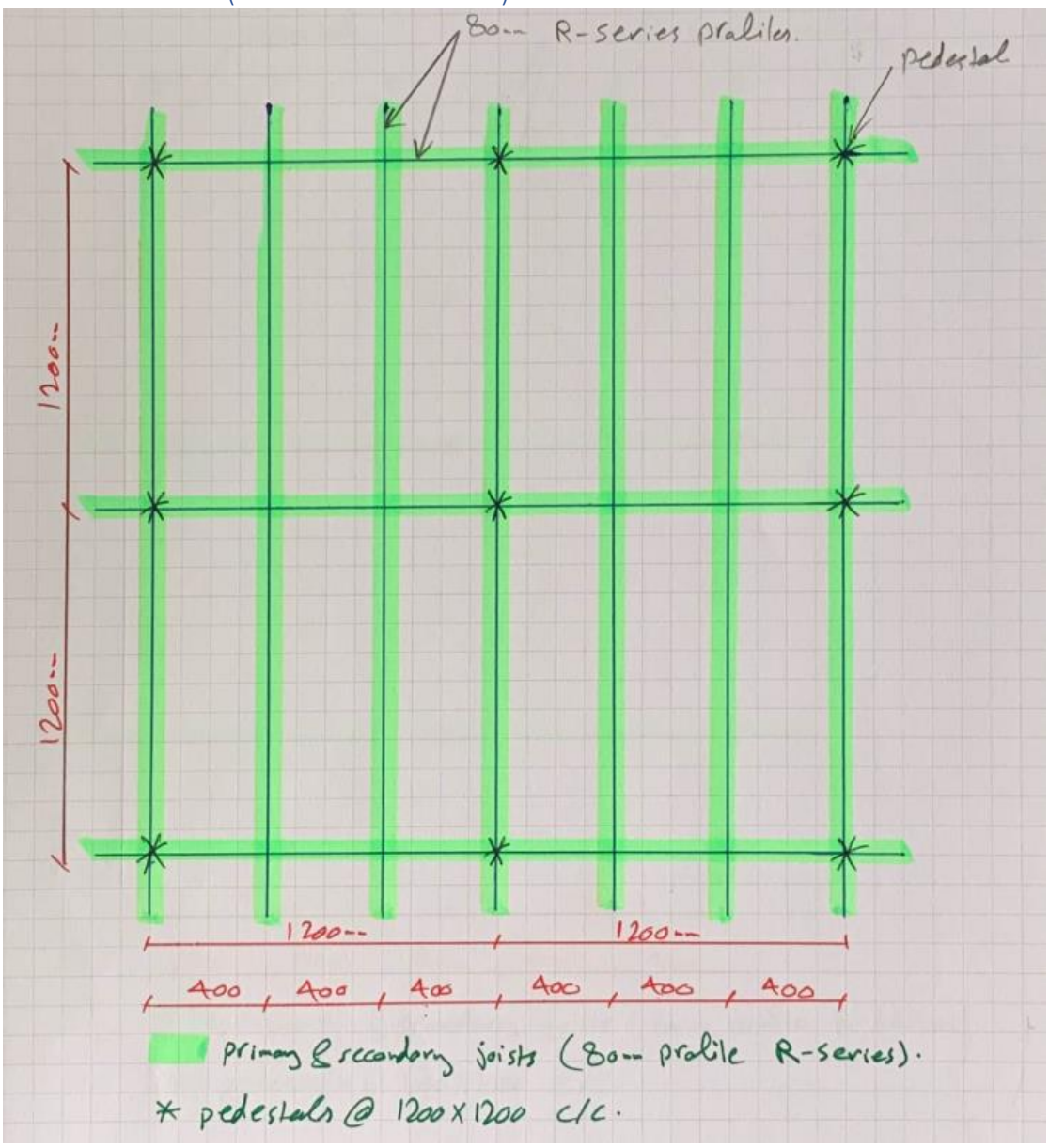
$$\alpha_\text{H} = \frac{5\text{w}\text{l}^4}{384\text{EI}}$$

$$\alpha_\text{H} = \frac{5\times3.642\times1300^4}{384\times70000\times455800} = 4.3\text{mm} < 6.5\text{mm} \rightarrow \left(\frac{1300}{200}\right)$$ **Okay**

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 37
Date: 09/04/2021	**By:** A.N

Aluminium Profile Design 80mm X 80mm Profile Grid (80mm Profile R Series):

Sketch Of Grid Line System For the Last Case Scenario for 3kN/m2 Live load and Dead load of the Pavers (80mm Profile R Series):



Aluminium Profile Checked for Live load of 3kN/m2 and Pavers for the Last Case Scenario (80mm Profile R Series):

The Profile has the following particulars:

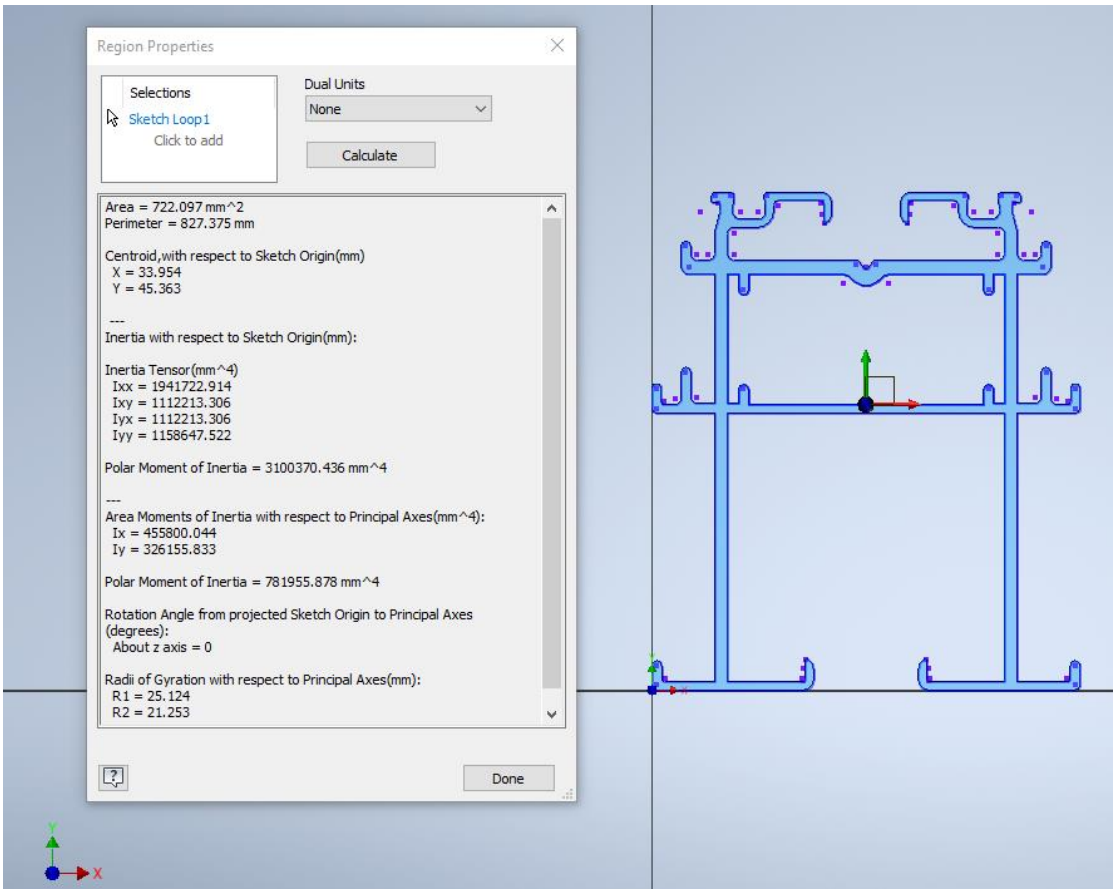
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 455800mm^4$ (80mm R Series Profile)

$Z = 10048mm^3$ (80mm R Series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



80mm Profiles at Every 1200mm C/C.

Span between pedestals is 1200mm C/C (Profile Length)

Live Load:

UDL Live load $= 3\text{kN/m}^2 \times 1.2\text{m} = 3.6\text{kN/m}$ (SLS)

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 39
Date: 09/04/2021	**By:** A.N

Dead Load for Pavers:

Dimensions (mm)	Colours	Pcs per m2	m2 per bale	Pcs per Bale	No. Layers	Approx. Bale Weight (KG)	Profile	Strength
400 x 400 x 40	Silver Granite, Black Granite, Buff Granite	6.25	13.44	84	14	1215	Chamfer	Class 1
450 x 450 x 40	Silver Granite, Black Granite, Buff Granite,	4.94	12.96	64	16	1150	Square edge	Class 1
400 x 400 x 50	Silver Granite, Black Granite, Buff Granite	6.25	11.52	72	12	1215	Chamfer	Class 1

Dead load due to Pavers: Convert from kg to kN $\rightarrow 1215\text{kg} = 11.92\text{kN}$

Dead load due to Pavers: if bale area 11.52m^2 gave us bale weight of 11.92kN,
Therefore $\rightarrow \left(\frac{11.92\text{kN}}{11.52\text{m}^2}\right) = 1.03\text{kN/m}^2$

UDL Dead load due to Pavers: $1.03\text{kN/m}^2 \times 1.2\text{m} = 1.236\text{kN/m}$

Dead Load For 80mm Profiles spanning 1200mm:

Dead load due to secondary Joists: $2\text{Nr} \times (0.000722\text{m}^2) \times 27\text{kN/m}^3 = 0.039\text{kN/m}$

Total UDL Load: $1.236 + 0.039 = 1.275\text{kN/m}$ (SLS)

Total UDL Load: $(3.6\text{kN/m} \times 1.5) + (1.275\text{kN/m} \times 1.35) = 7.12\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{7.12 \times 1.2^2}{8} = 1.28\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{1.28 \times 10^6}{10048} = 127.4\frac{N}{mm^2} < 130\frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

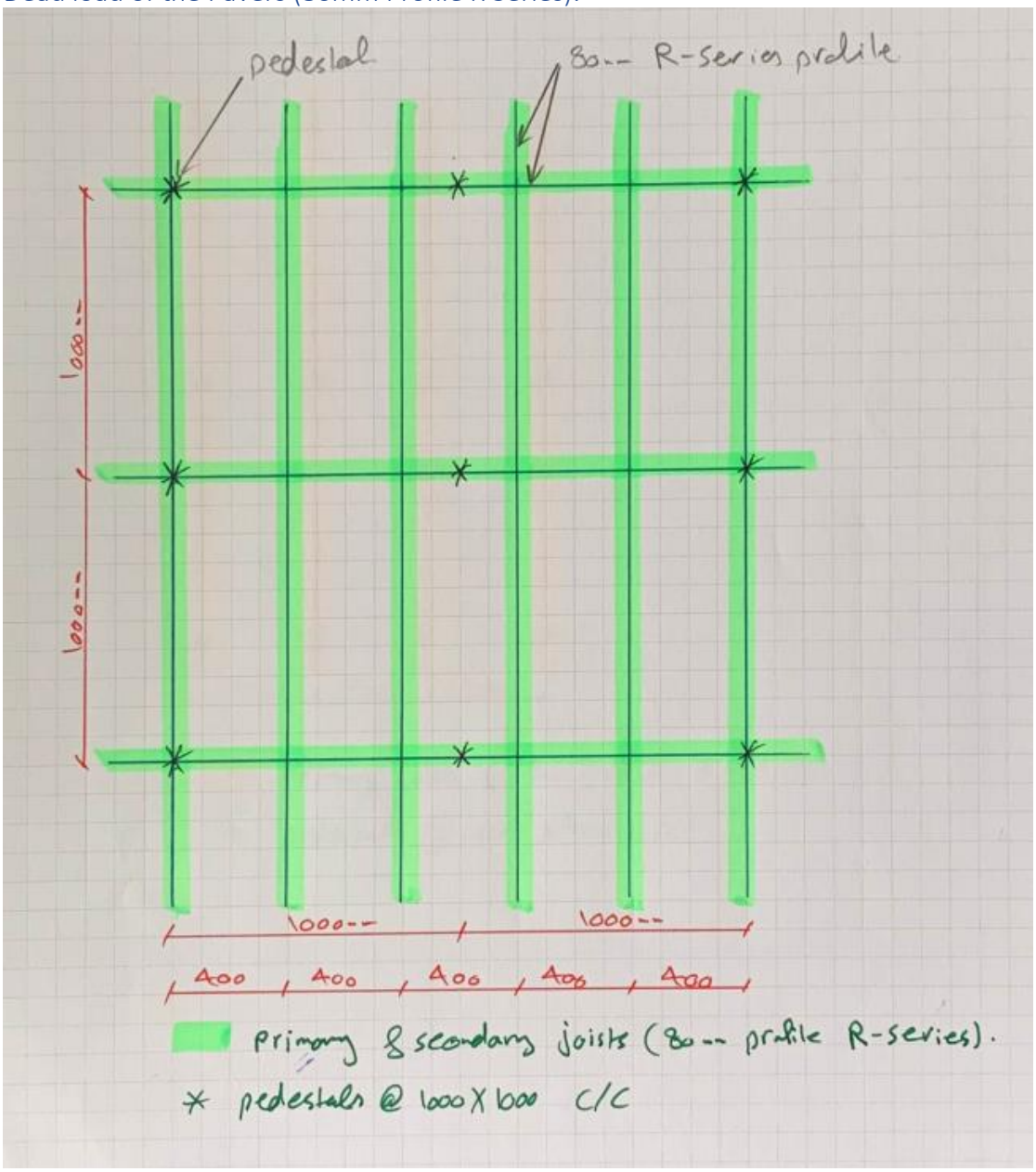
$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 4.875 \times 1200^4}{384 \times 70000 \times 455800} = 4.13\text{mm} < 6\text{mm} \rightarrow \left(\frac{1200}{200}\right)$$ **Okay**

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 40
Date: 09/04/2021	**By:** A.N

Sketch Of Grid Line System For the Additional Case Scenario for 5kN/m2 Live load and Dead load of the Pavers (80mm Profile R Series):



Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Pavers for the Additional Case Scenario (80mm Profile R Series):

The Profile has the following particulars:

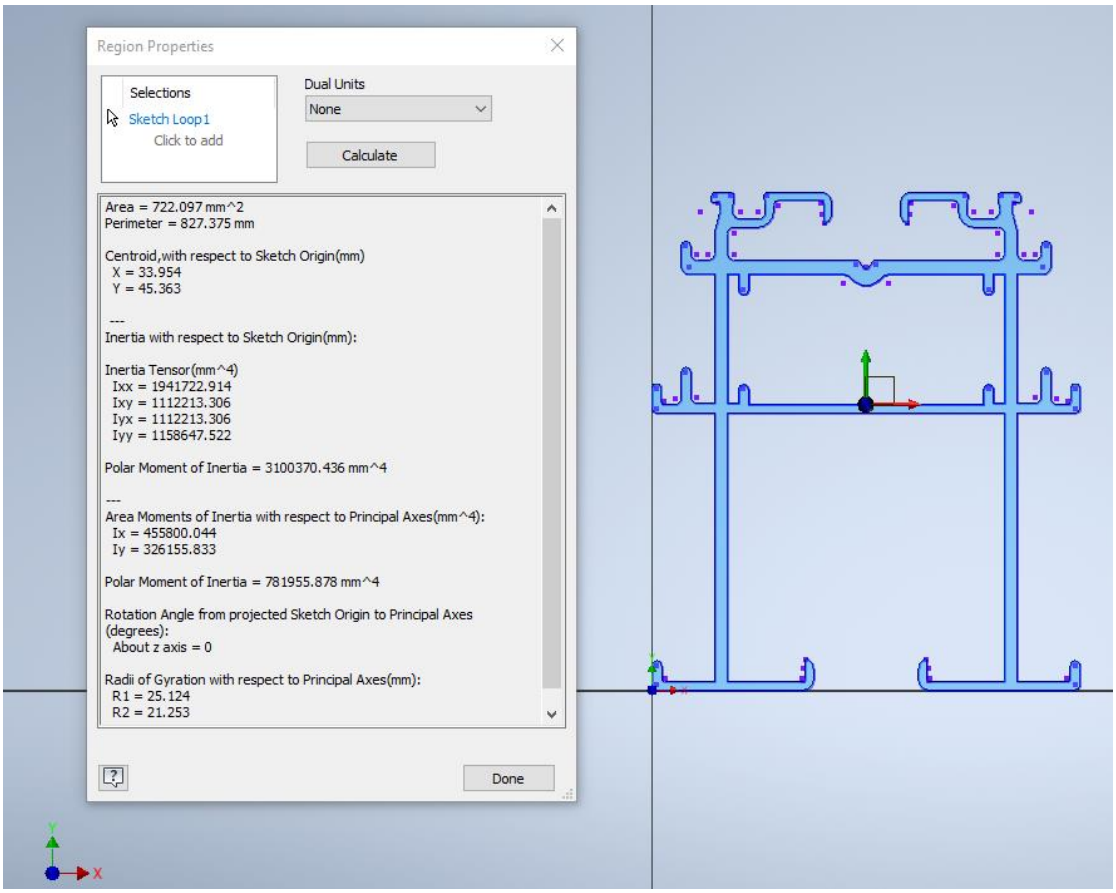
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 455800mm^4$ (80mm R Series Profile)

$Z = 10048mm^3$ (80mm R Series Profile)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



80mm Profiles at Every 1000mm C/C.

Span between pedestals is 1000mm C/C (Profile Length)

Live Load:

$$\text{UDL Live load} = 5\text{kN/m}^2 \times 1\text{m} = 5\text{kN/m}$$

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 42
Date: 09/04/2021	**By:** A.N

Dead Load for Pavers:

Dimensions (mm)	Colours	Pcs per m2	m2 per bale	Pcs per Bale	No. Layers	Approx. Bale Weight (KG)	Profile	Strength
400 x 400 x 40	Silver Granite, Black Granite, Buff Granite	6.25	13.44	84	14	1215	Chamfer	Class 1
450 x 450 x 40	Silver Granite, Black Granite, Buff Granite,	4.94	12.96	64	16	1150	Square edge	Class 1
400 x 400 x 50	Silver Granite, Black Granite, Buff Granite	6.25	11.52	72	12	1215	Chamfer	Class 1

Dead load due to Pavers: Convert from kg to kN $\rightarrow 1215\text{kg} = 11.92\text{kN}$

Dead load due to Pavers: if bale area 11.52m^2 gave us bale weight of 11.92kN,
Therefore $\rightarrow \left(\frac{11.92\text{kN}}{11.52\text{m}^2}\right) = 1.03\text{kN/m}^2$

UDL Dead load due to Pavers: $1.03\text{kN/m}^2 \times 1\text{m} = 1.03\text{kN/m}$

Dead Load For 80mm Profiles spanning 1000mm:

Dead load due to secondary Joists: $2\text{Nr} \times (0.000722\text{m}^2) \times 27\text{kN/m}^3 = 0.039\text{kN/m}$

Total UDL Load: $1.03 + 0.039 = 1.069\text{kN/m}$ (SLS)

Total UDL Load: $(5\text{kN/m} \times 1.5) + (1.069\text{kN/m} \times 1.35) = 8.94\text{kN/m}$ (ULS)

Maximum Moment:

$$\text{M}_\text{H} = \frac{\text{w} \times \text{l}^2}{8}$$

$$\text{M}_\text{H} = \frac{8.94 \times 1^2}{8} = 1.12\text{kNm}$$

Maximum Stress:

$$\sigma_{\max} = \frac{\text{M}}{\text{Z}}$$

$$\sigma_{\max} = \frac{1.12 \times 10^6}{10048} = 112\,\frac{\text{N}}{\text{mm}^2} < 130\,\frac{\text{N}}{\text{mm}^2}$$ **Okay**

Maximum Deflection:

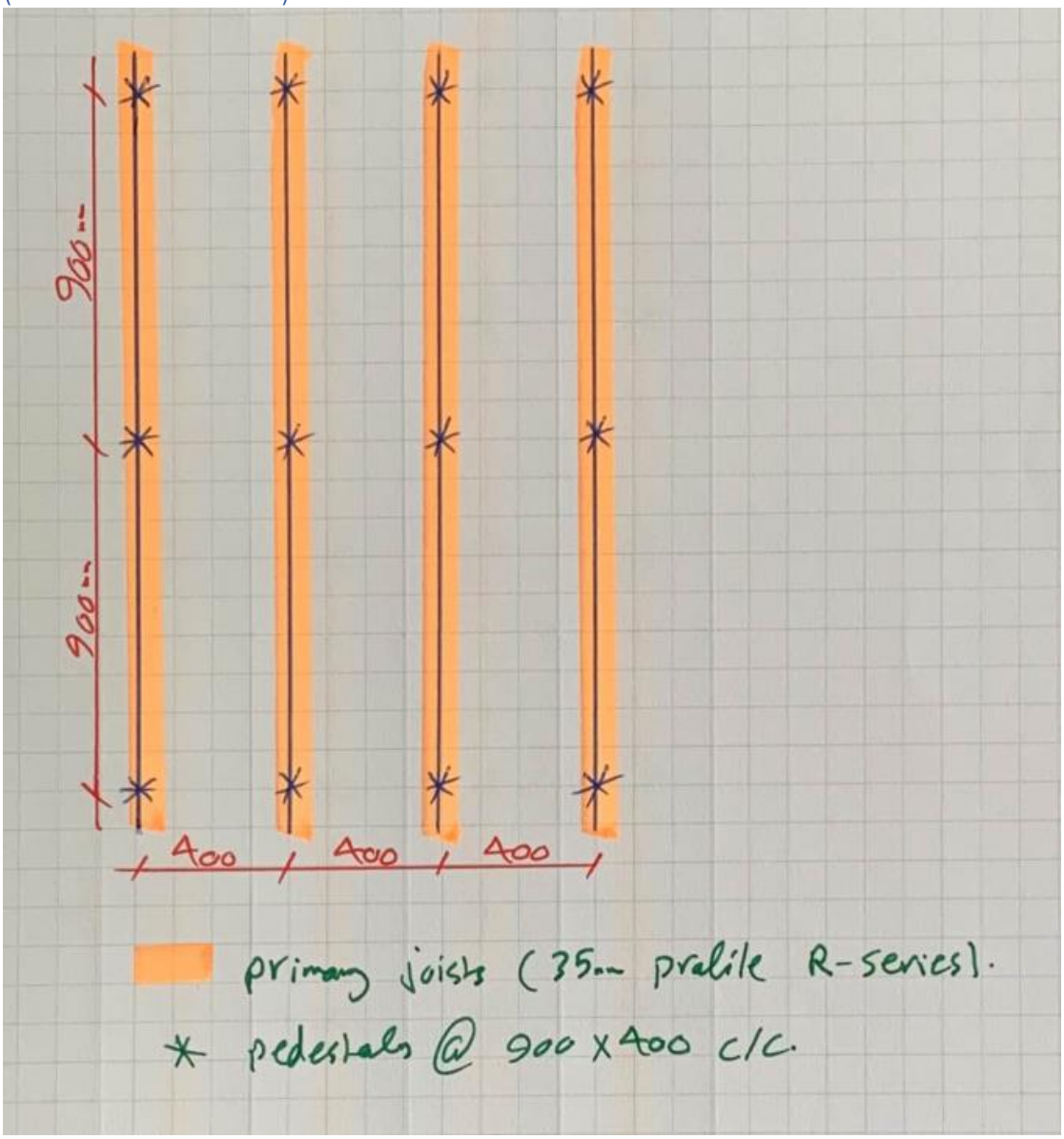
$$\alpha_\text{H} = \frac{5\text{w}\text{l}^4}{384\text{EI}}$$

$$\alpha_\text{H} = \frac{5 \times 6.069 \times 1000^4}{384 \times 70000 \times 455800} = 2.5\text{mm} < 5\text{mm} \rightarrow \left(\frac{1000}{200}\right)$$ **Okay**

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 43
Date: 09/04/2021	**By:** A.N

Aluminium Profile Design For the General Case Scenario (35mm Profile R Series):

Sketch Of Grid Line System For the General Case Scenario For 3kN/m2 Live load (35mm Profile R Series):



Aluminium Profile Checked for Live load of 3kN/m2 and Dead Load of Pavers for the General Case Scenario (35mm Profile R Series):

The Profile has the following particulars:

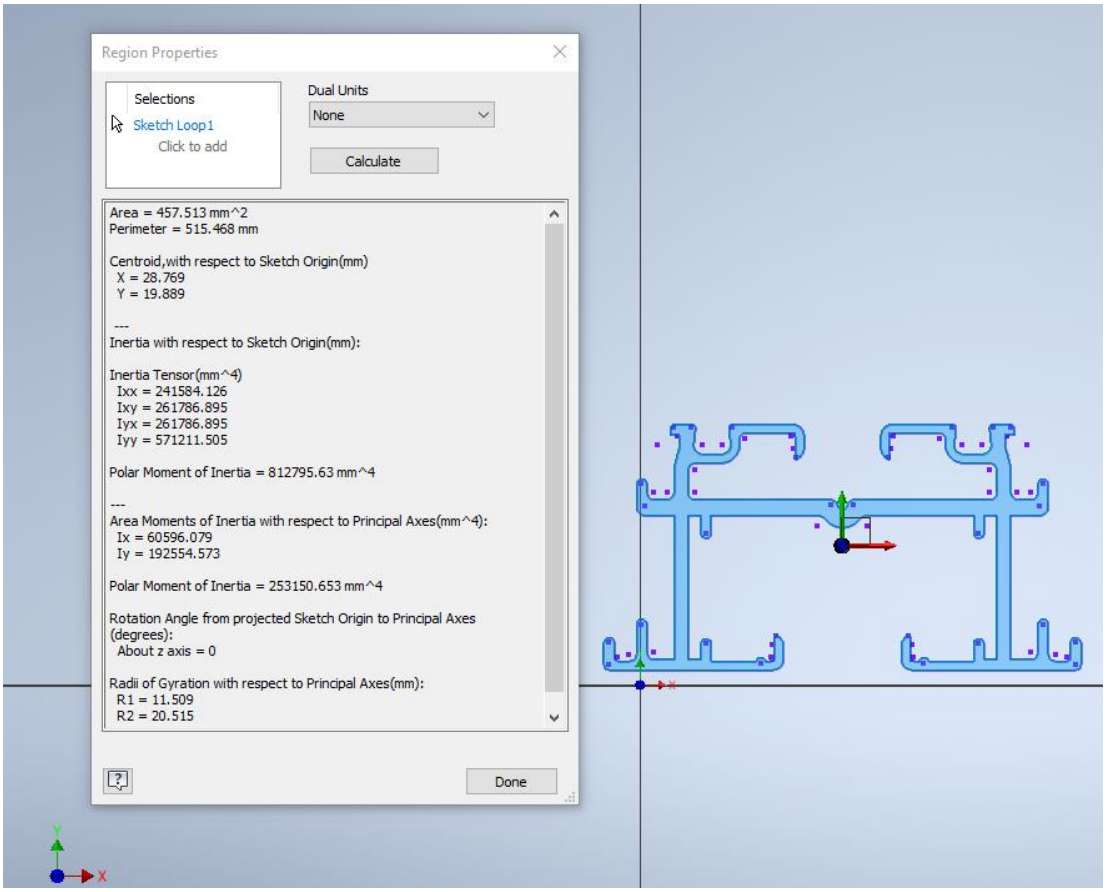
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 60596mm^4$ (35mm Profile R - Series)

$Z = 3047mm^3$ (35mm Profile R - Series)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



35mm Profiles R Series at Every 400mm C/C.

35mm Profile R Series Span is 900mm (Profile Length)

Live Load:

UDL Live load $= 3\text{kN/m}^2 \times 0.4\text{m} = 1.2\text{kN/m}$ (SLS)

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 45
Date: 09/04/2021	**By:** A.N

Dead Load for Pavers:

Dimensions (mm)	Colours	Pcs per m2	m2 per bale	Pcs per Bale	No. Layers	Approx. Bale Weight (KG)	Profile	Strength
400 x 400 x 40	Silver Granite, Black Granite, Buff Granite	6.25	13.44	84	14	1215	Chamfer	Class 1
450 x 450 x 40	Silver Granite, Black Granite, Buff Granite,	4.94	12.96	64	16	1150	Square edge	Class 1
400 x 400 x 50	Silver Granite, Black Granite, Buff Granite	6.25	11.52	72	12	1215	Chamfer	Class 1

Dead load due to Pavers: Convert from kg to kN $\rightarrow 1215\text{kg} = 11.92\text{kN}$

Dead load due to Pavers: if bale area 11.52m^2 gave us bale weight of 11.92kN,
Therefore $\rightarrow \left(\frac{11.92\text{kN}}{11.52\text{m}^2}\right) = 1.03\text{kN/m}^2$

UDL Dead load due to Pavers: $1.03\text{kN/m}^2 \times 0.4\text{m} = 0.412\text{kN/m}$

Total UDL Load: $(1.2\text{kN/m} \times 1.5) + (0.412\text{kN/m} \times 1.35) = 2.356\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{2.356 \times 0.9^2}{8} = 0.24\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{0.24 \times 10^6}{3047} = 79\frac{N}{mm^2} < 130\frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

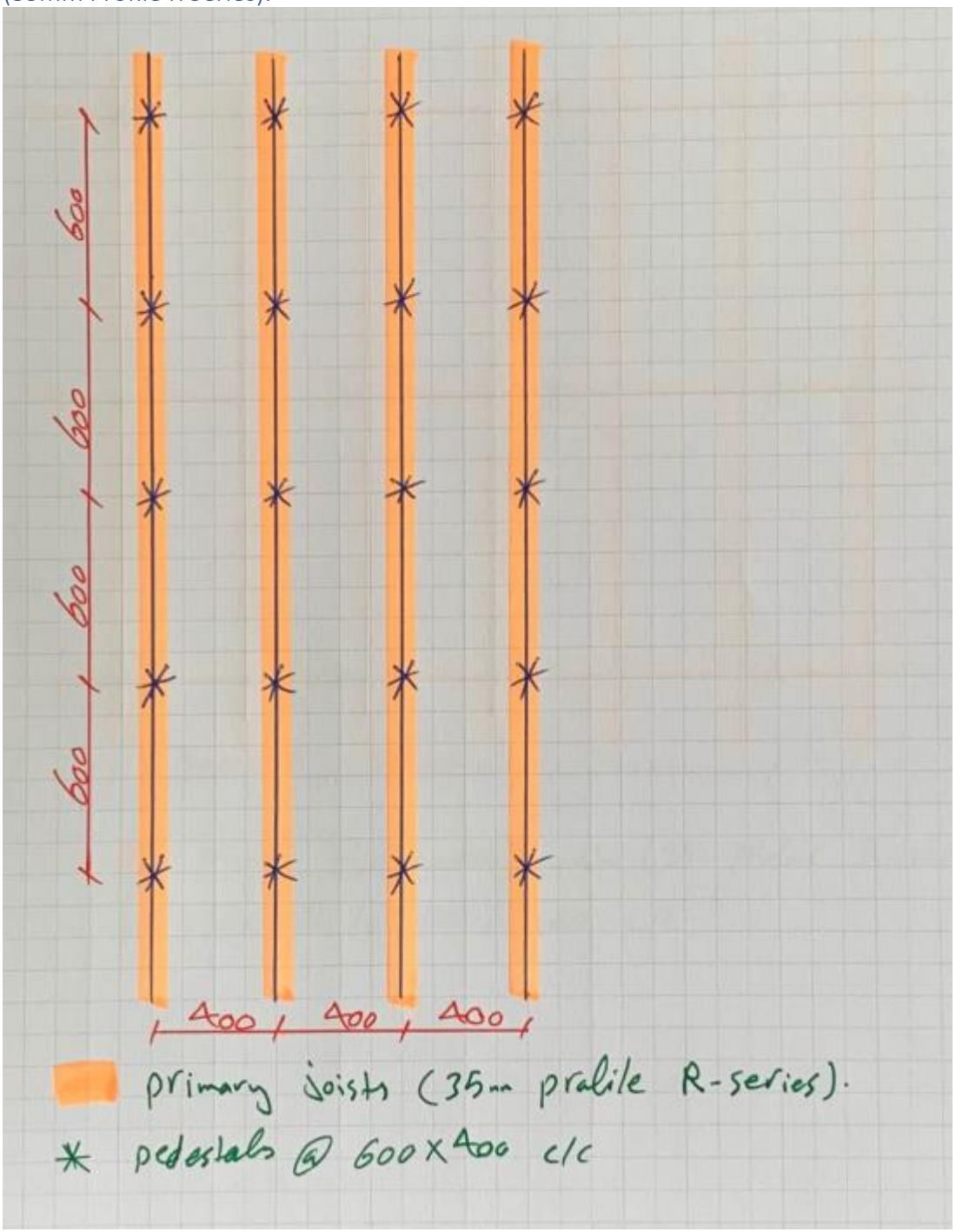
$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 1.612 \times 900^4}{384 \times 70000 \times 60596} = 3.25\text{mm} < 4.5\text{mm} \rightarrow \left(\frac{900}{200}\right)$$ **Okay**

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Project: Aluminium Profile	Contract: 1513-1
Subject: Aluminium Profile Design	Sheet No. 46
Date: 09/04/2021	By: A.N

Sketch Of Grid Line System For the General Case Scenario For 5kN/m2 Live load (35mm Profile R Series):



Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Pavers for the General Case Scenario (35mm Profile R Series):

The Profile has the following particulars:

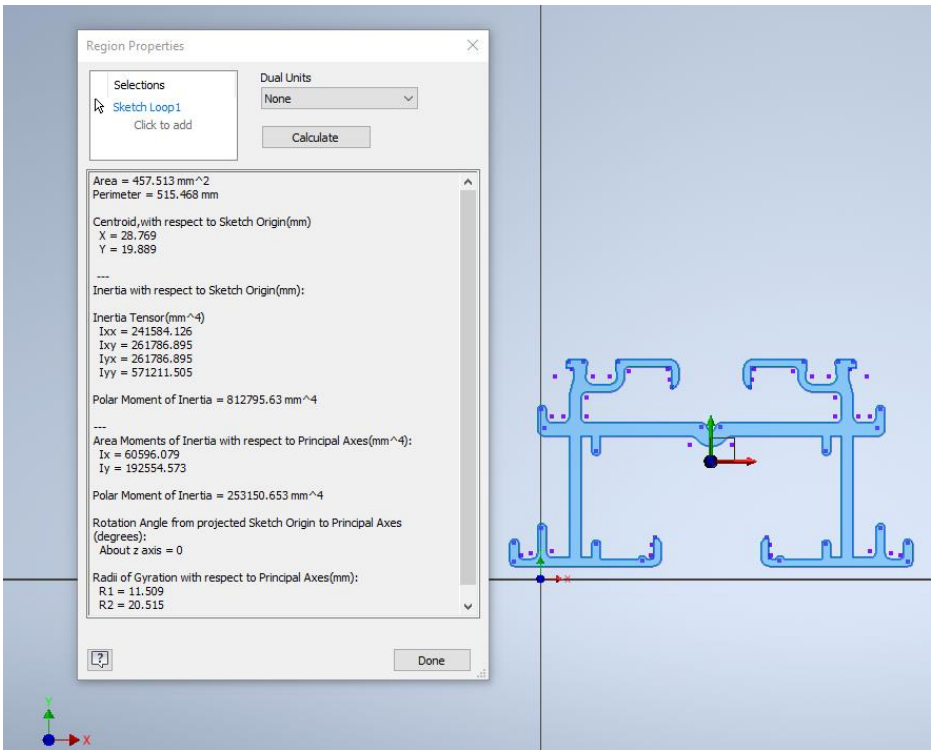
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 60596mm^4$ (35mm Profile R - Series)

$Z = 3047mm^3$ (35mm Profile R - Series)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



Span Between 80mm Profiles = 400mm

Span Between pedestals = 600mm (Profile Length)

Live Load:

$\text{UDL Live load} = 5\text{kN/m}^2 \times 0.4\text{m} = 2\text{kN/m}$

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 48
Date: 09/04/2021	**By:** A.N

Dead Load for Pavers:

Dimensions (mm)	Colours	Pcs per m2	m2 per bale	Pcs per Bale	No. Layers	Approx. Bale Weight (KG)	Profile	Strength
400 x 400 x 40	Silver Granite, Black Granite, Buff Granite	6.25	13.44	84	14	1215	Chamfer	Class 1
450 x 450 x 40	Silver Granite, Black Granite, Buff Granite,	4.94	12.96	64	16	1150	Square edge	Class 1
400 x 400 x 50	Silver Granite, Black Granite, Buff Granite	6.25	11.52	72	12	1215	Chamfer	Class 1

Dead load due to Pavers: Convert from kg to kN $\rightarrow 1215\text{kg} = 11.92\text{kN}$

Dead load due to Pavers: if bale area 11.52m^2 gave us bale weight of 11.92kN,
Therefore $\rightarrow \left(\frac{11.92\text{kN}}{11.52\text{m}^2}\right) = 1.03\text{kN/m}^2$

UDL Dead load due to Pavers: $1.03\text{kN/m}^2 \times 0.4\text{m} = 0.412\text{kN/m}$

Total UDL Load: $(2\text{kN/m} \times 1.5) + (0.412\text{kN/m} \times 1.35) = 3.56\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{3.56 \times 0.6^2}{8} = 0.16\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{0.16 \times 10^6}{3047} = 53\,\frac{N}{mm^2} < 130\,\frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

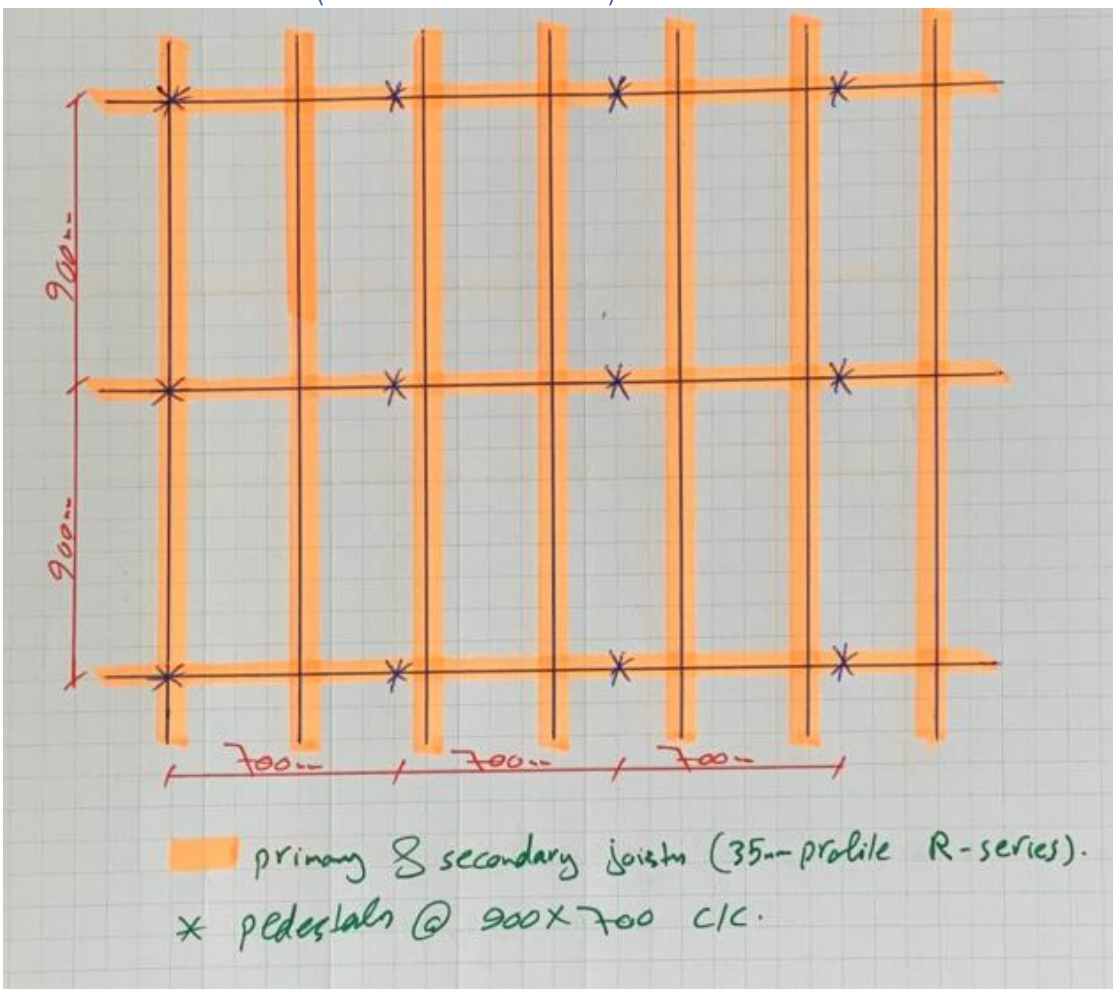
$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 2.412 \times 600^4}{384 \times 70000 \times 60596} = 1\text{mm} < 3\text{mm} \rightarrow \left(\frac{600}{200}\right)$$ **Okay**

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 49
Date: 09/04/2021	**By:** A.N

Aluminium Profile Design For the Occasional Case Scenario (35mm Profile R Series):

Sketch Of Grid Line System For the Occasional Case Scenario for 3kN/m2 Live load and Dead load of the Pavers (35mm Profile R Series):



Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 50
Date: 09/04/2021	**By:** A.N

Aluminium Profile Checked for Live load of 3kN/m2 and Deal load of the pavers for the Occasional Case Scenario (35mm Profile R Series):

The Profile has the following particulars:

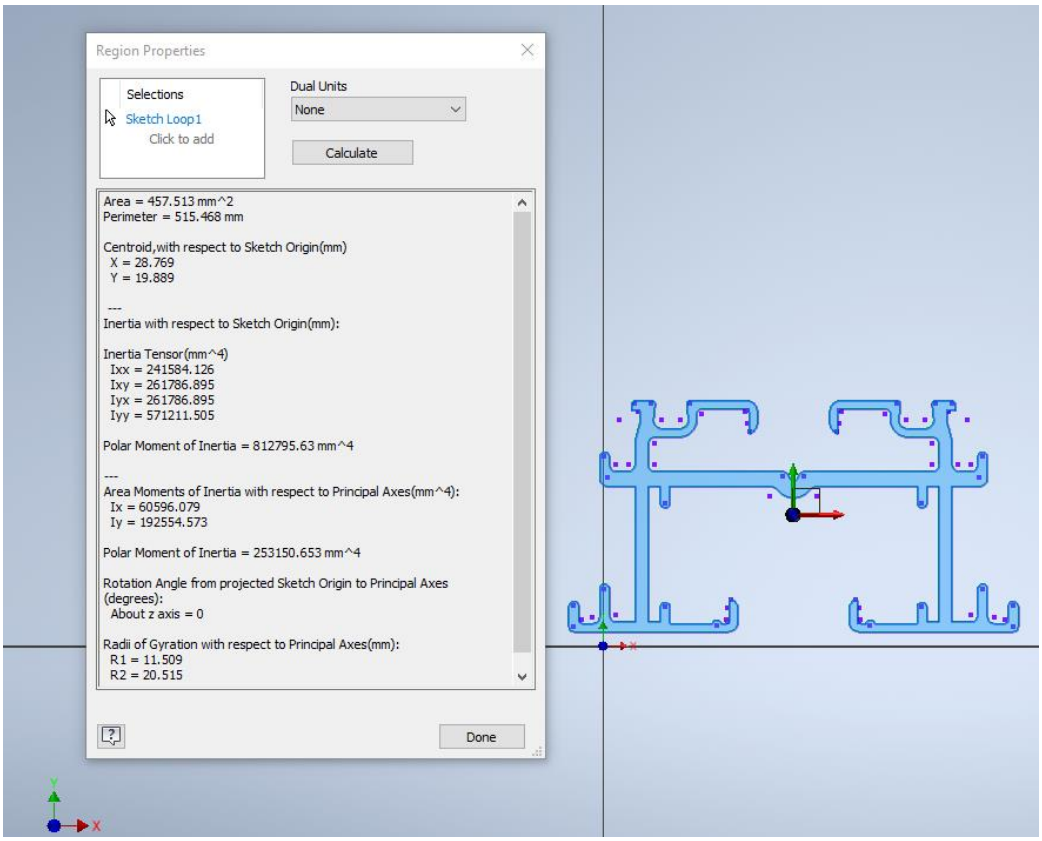
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 60596mm^4$ (35mm Profile R - Series)

$Z = 3047mm^3$ (35mm Profile R - Series)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



Live Load:

UDL Live load $= 3\text{kN/m}^2 \times 0.9\text{m} = 2.7\text{kN/m}$ (SLS)

Dead Load Of Secondary Joists:

Dead load due to secondary Joists: $2\text{Nr} \times 0.000457\text{m}^2 \times 27\text{kN/m}^3 = 0.024\text{kN/m}$ (SLS)

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 51
Date: 09/04/2021	**By:** A.N

Dead Load for Pavers:

Dimensions (mm)	Colours	Pcs per m2	m2 per bale	Pcs per Bale	No. Layers	Approx. Bale Weight (KG)	Profile	Strength
400 x 400 x 40	Silver Granite, Black Granite, Buff Granite	6.25	13.44	84	14	1215	Chamfer	Class 1
450 x 450 x 40	Silver Granite, Black Granite, Buff Granite,	4.94	12.96	64	16	1150	Square edge	Class 1
400 x 400 x 50	Silver Granite, Black Granite, Buff Granite	6.25	11.52	72	12	1215	Chamfer	Class 1

Dead load due to Pavers: Convert from kg to kN $\rightarrow 1215\text{kg} = 11.92\text{kN}$

Dead load due to Pavers: if bale area 11.52m^2 gave us bale weight of 11.92kN,
Therefore $\rightarrow \left(\frac{11.92\text{kN}}{11.52\text{m}^2}\right) = 1.03\text{kN/m}^2$ (SLS)

UDL Dead load due to Pavers: $1.03\text{kN/m}^2 \times 0.9\text{m} = 0.927\text{kN/m}$

Total UDL Dead Load: $(0.927\text{kN/m} + 0.024\text{kN/m}) = 0.95\text{kN/m}$ (SLS)

Total UDL Load: $(2.7\text{kN/m} \times 1.5) + (0.95\text{kN/m} \times 1.35) = 5.33\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{5.33 \times 0.7^2}{8} = 0.326\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{0.326 \times 10^6}{3047} = 107\,\frac{N}{mm^2} < 130\,\frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

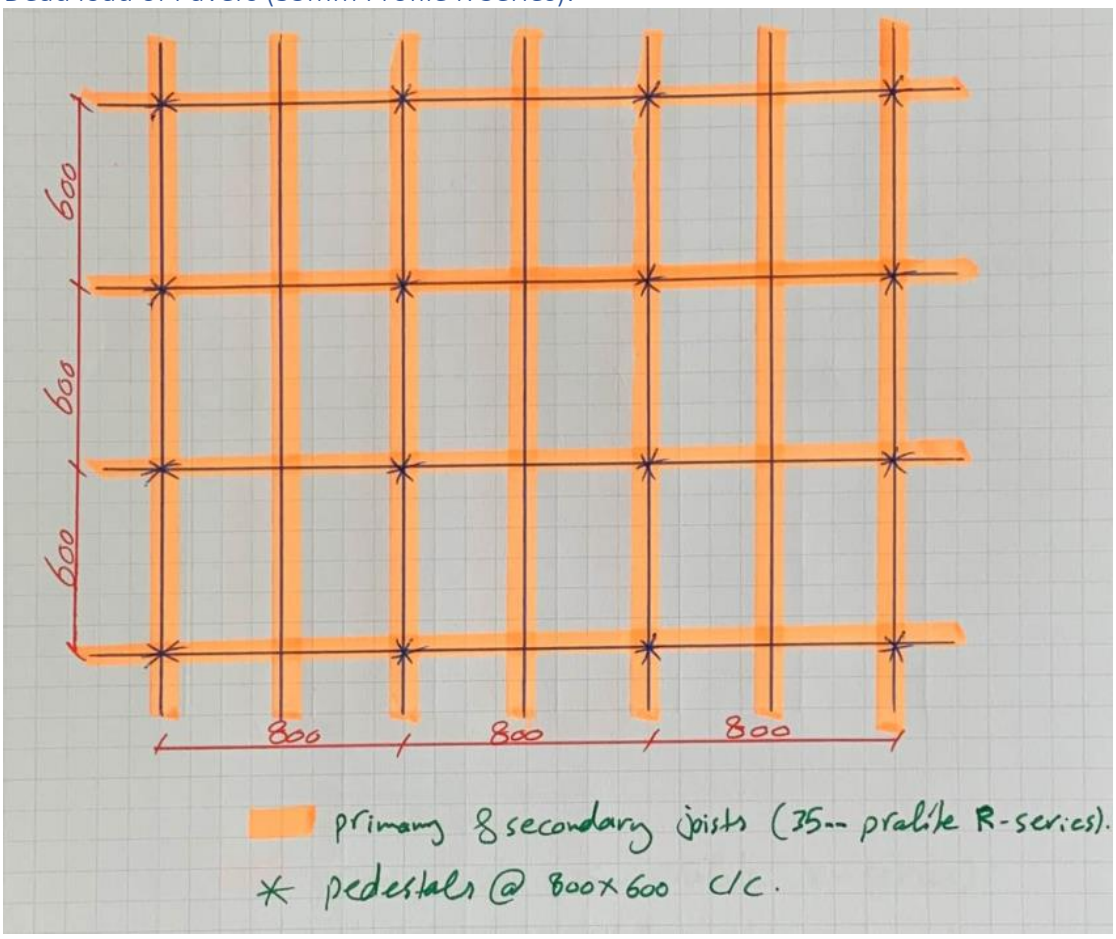
$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 3.651 \times 700^4}{384 \times 70000 \times 60596} = 2.7\text{mm} < 3.5\text{mm} \rightarrow \left(\frac{700}{200}\right)$$ **Okay**

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 52
Date: 09/04/2021	**By:** A.N

Sketch Of Grid Line System For the Occasional Case Scenario for 5kN/m2 Live load and Dead load of Pavers (35mm Profile R Series):



Aluminium Profile Checked for Live load of 5kN/m2 and Dead Load of the Pavers for the Occasional Case Scenario (35mm Profile R Series):

The Profile has the following particulars:

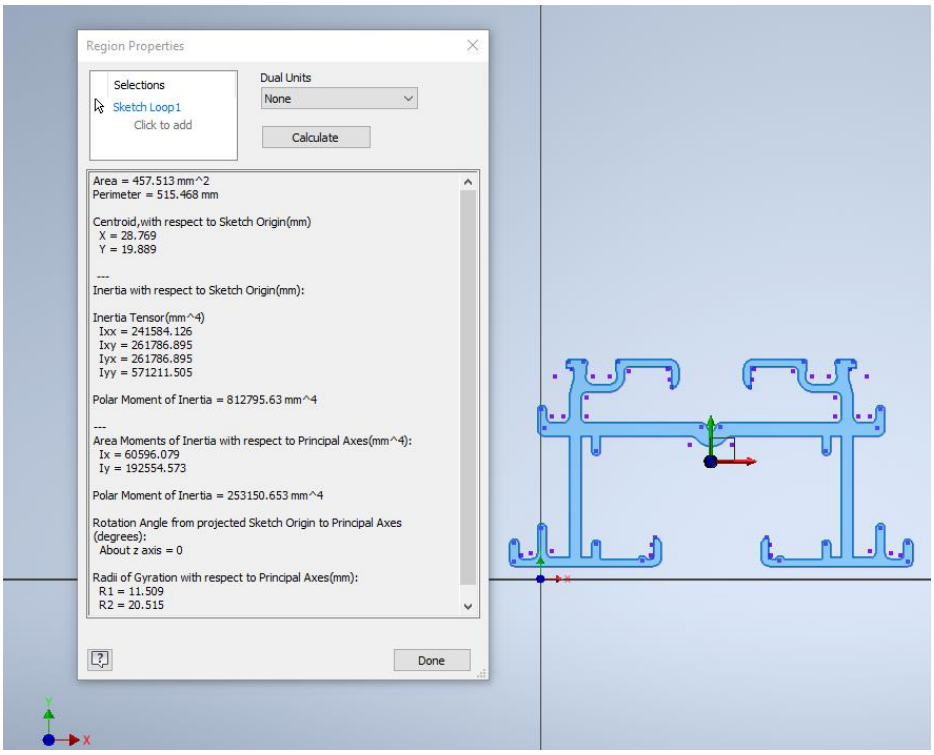
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 60596mm^4$ (35mm Profile R - Series)

$Z = 3047mm^3$ (35mm Profile R - Series)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



Live Load:

UDL Live load $= 3\text{kN/m}^2 \times 0.6\text{m} = 1.8\text{kN/m}$ (SLS)

Dead Load Of Secondary Joists:

Assuming the area of secondary profiles are half of the primary ones:

Dead load due to secondary Joists: $2\text{Nr} \times 0.000457\text{m}^2 \times 27\text{kN/m}^3 = 0.024\text{kN/m}$ (SLS)

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 54
Date: 09/04/2021	**By:** A.N

Dead Load for Pavers:

Dimensions (mm)	Colours	Pcs per m2	m2 per bale	Pcs per Bale	No. Layers	Approx. Bale Weight (KG)	Profile	Strength
400 x 400 x 40	Silver Granite, Black Granite, Buff Granite	6.25	13.44	84	14	1215	Chamfer	Class 1
450 x 450 x 40	Silver Granite, Black Granite, Buff Granite,	4.94	12.96	64	16	1150	Square edge	Class 1
400 x 400 x 50	Silver Granite, Black Granite, Buff Granite	6.25	11.52	72	12	1215	Chamfer	Class 1

Dead load due to Pavers: Convert from kg to kN $\rightarrow 1215\text{kg} = 11.92\text{kN}$

Dead load due to Pavers: if bale area 11.52m^2 gave us bale weight of 11.92kN,
Therefore $\rightarrow \left(\frac{11.92\text{kN}}{11.52\text{m}^2}\right) = 1.03\text{kN/m}^2$

UDL Dead load due to Pavers: $1.03\text{kN/m}^2 \times 0.6\text{m} = 0.618\text{kN/m}$

Total UDL Dead Load: $(0.618\text{kN/m} + 0.024\text{kN/m}) = 0.642\text{kN/m}$ (SLS)

Total UDL Load: $(1.8\text{kN/m} \times 1.5) + (0.642\text{kN/m} \times 1.35) = 3.57\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{3.57 \times 0.8^2}{8} = 0.285\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{0.285 \times 10^6}{3047} = 94\,\frac{N}{mm^2} < 130\,\frac{N}{mm^2}$$ **Okay**

Maximum Deflection:

$$\alpha_H = \frac{5wl^4}{384EI}$$

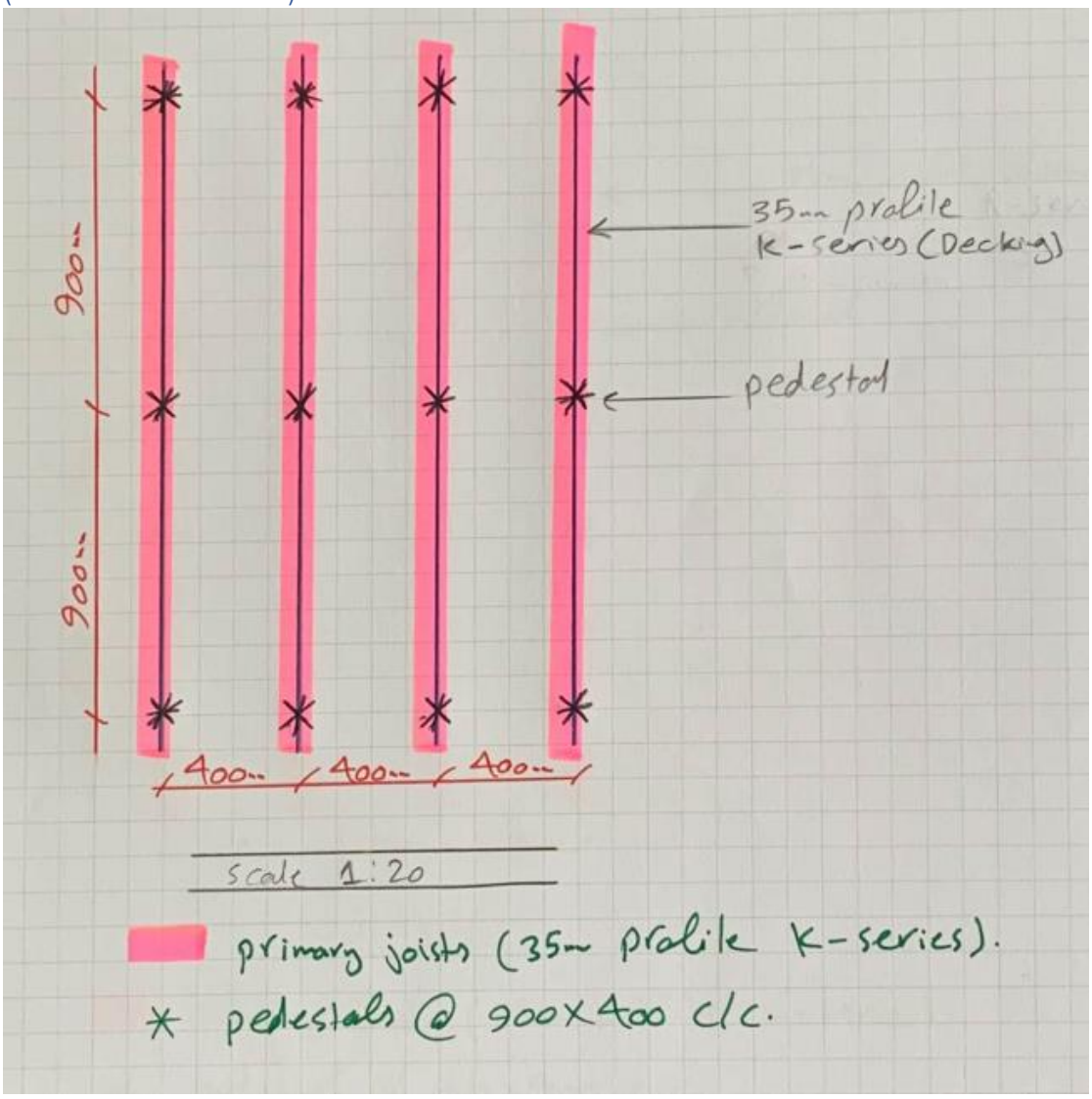
$$\alpha_H = \frac{5 \times 2.442 \times 800^4}{384 \times 70000 \times 60596} = 3.07\text{mm} < 4\text{mm} \rightarrow \left(\frac{800}{200}\right)$$ **Okay**

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Project: Aluminium Profile	Contract: 1513-1
Subject: Aluminium Profile Design	Sheet No. 55
Date: 09/04/2021	By: A.N

Aluminium Profile Design For the General Case Scenario (35mm Profile K Series):

Sketch Of Grid Line System For the General Case Scenario For 3kN/m2 Live load (35mm Profile K Series):



Aluminium Profile Checked for Live load of 3kN/m2 and Dead Load of Composite Decking for the General Case Scenario (35mm Profile K Series):

The Profile has the following particulars:

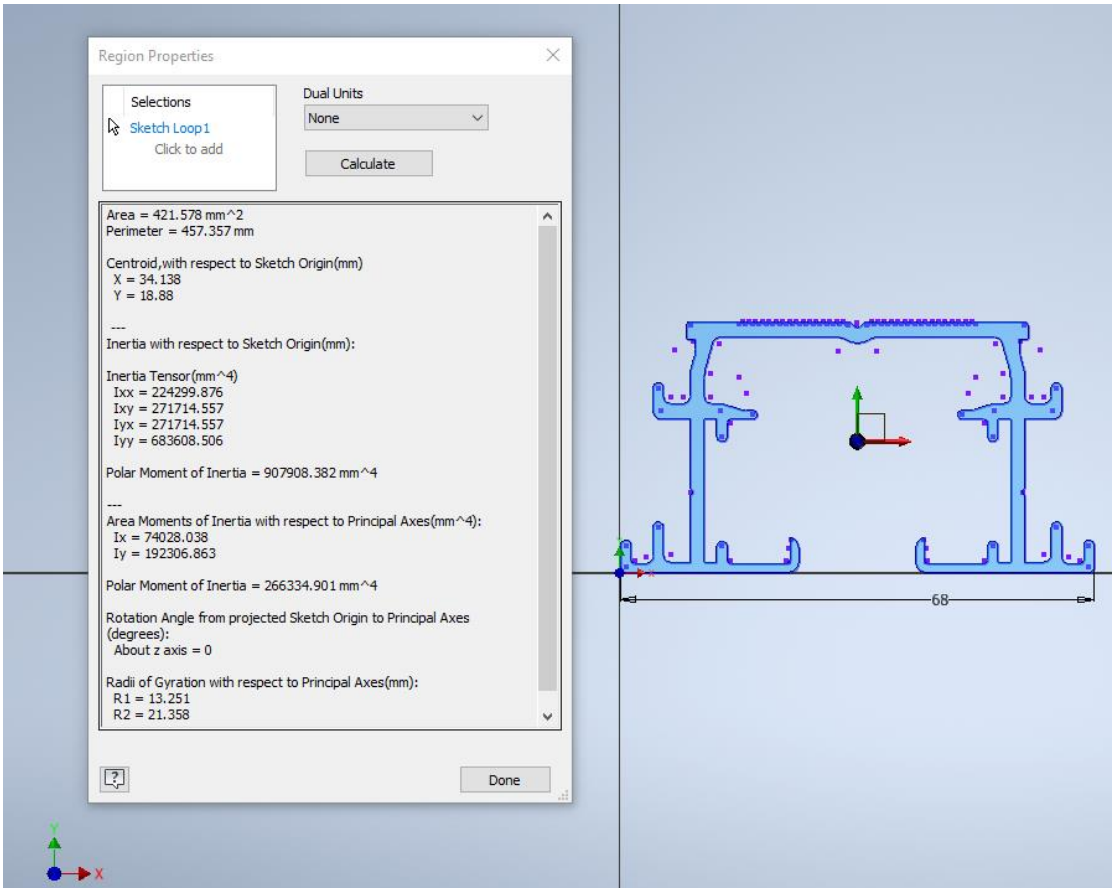
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 74028mm^4$ (35mm Profile K - Series)

$Z = 3920mm^3$ (35mm Profile K - Series)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



35mm Profiles K Series at Every 400mm C/C.

35mm Profile K Series Span is 900mm (Profile Length)

Live Load:

UDL Live load $= 3\text{kN/m}^2 \times 0.4\text{m} = 1.2\text{kN/m}$ (SLS)

Dead Load for Composite Deck:

Dead load due to Composite Deck → Convert $\frac{\text{lb}}{\text{m}}$ to $\frac{\text{lb}}{\text{m}^2} \rightarrow \frac{7.3\text{lb/m}}{0.136\text{m}} = 55.15\text{lb/m}^2$

Dead load due to Composite Deck → Convert $\frac{\text{lb}}{\text{m}^2}$ to $\frac{\text{kN}}{\text{m}^2} \rightarrow 55.15\text{lb/m}^2 = 0.245\text{kN/m}^2$

UDL Dead load due to Composite Deck: $0.245\text{kN/m}^2 \times 0.4\text{m} = 0.098\text{kN/m}$ (SLS)

Total UDL Load: $(1.2\text{kN/m} \times 1.5) + (0.098\text{kN/m} \times 1.35) = 1.9323\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{1.9323 \times 0.9^2}{8} = 0.196\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{0.196 \times 10^6}{3920} = 50\frac{\text{N}}{\text{mm}^2} < 130\frac{\text{N}}{\text{mm}^2}$$ **Okay**

Maximum Deflection:

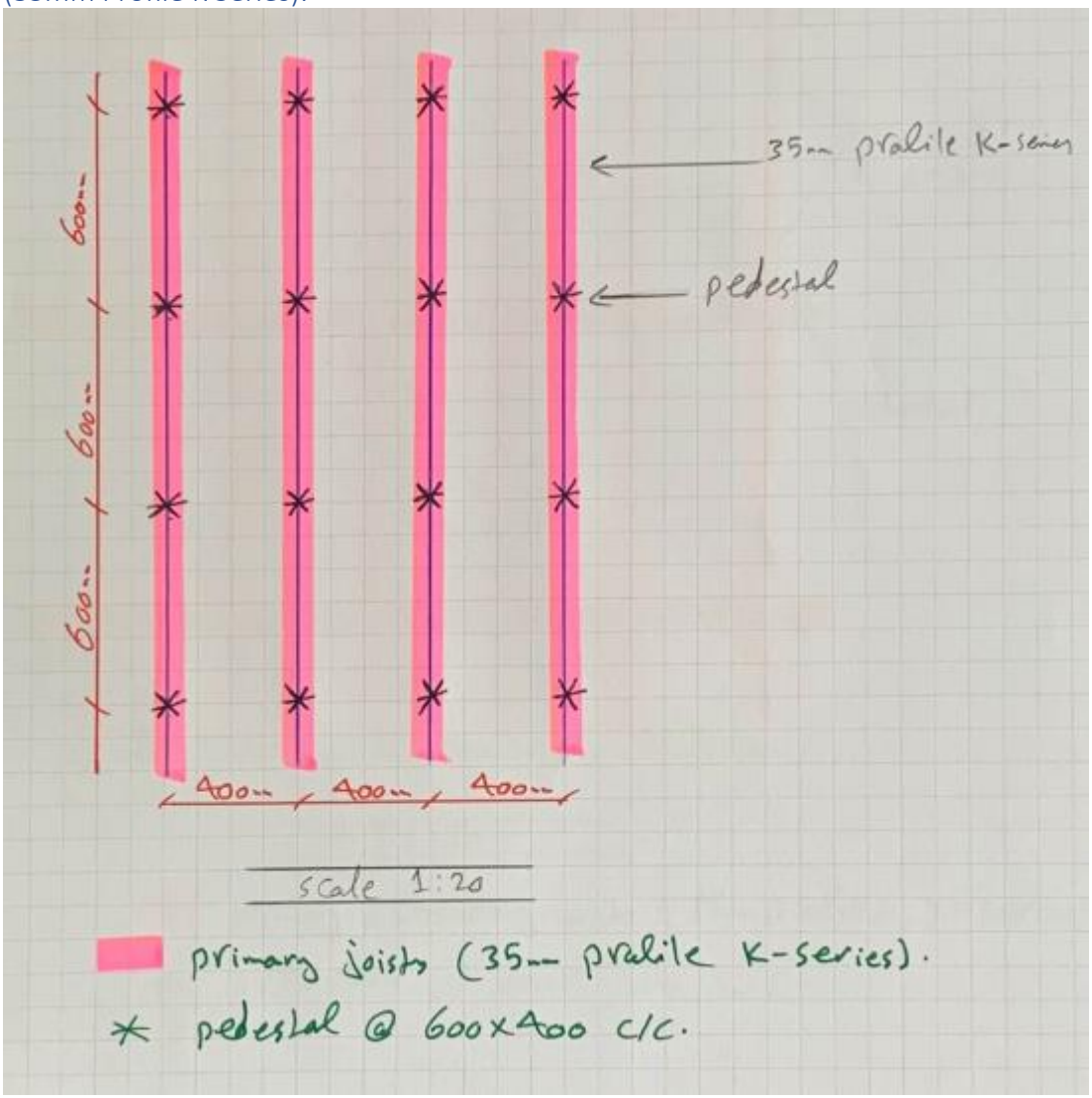
$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 1.298 \times 900^4}{384 \times 70000 \times 74028} = 2.14\text{mm} < 4.5\text{mm} \rightarrow \left(\frac{900}{200}\right)$$ **Okay**

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Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 58
Date: 09/04/2021	**By:** A.N

Sketch Of Grid Line System For the General Case Scenario For 5kN/m2 Live load (35mm Profile K Series):



Aluminium Profile Checked for Live load of 5kN/m2 and Dead load of Composite Decking for the General Case Scenario (35mm Profile K Series):

The Profile has the following particulars:

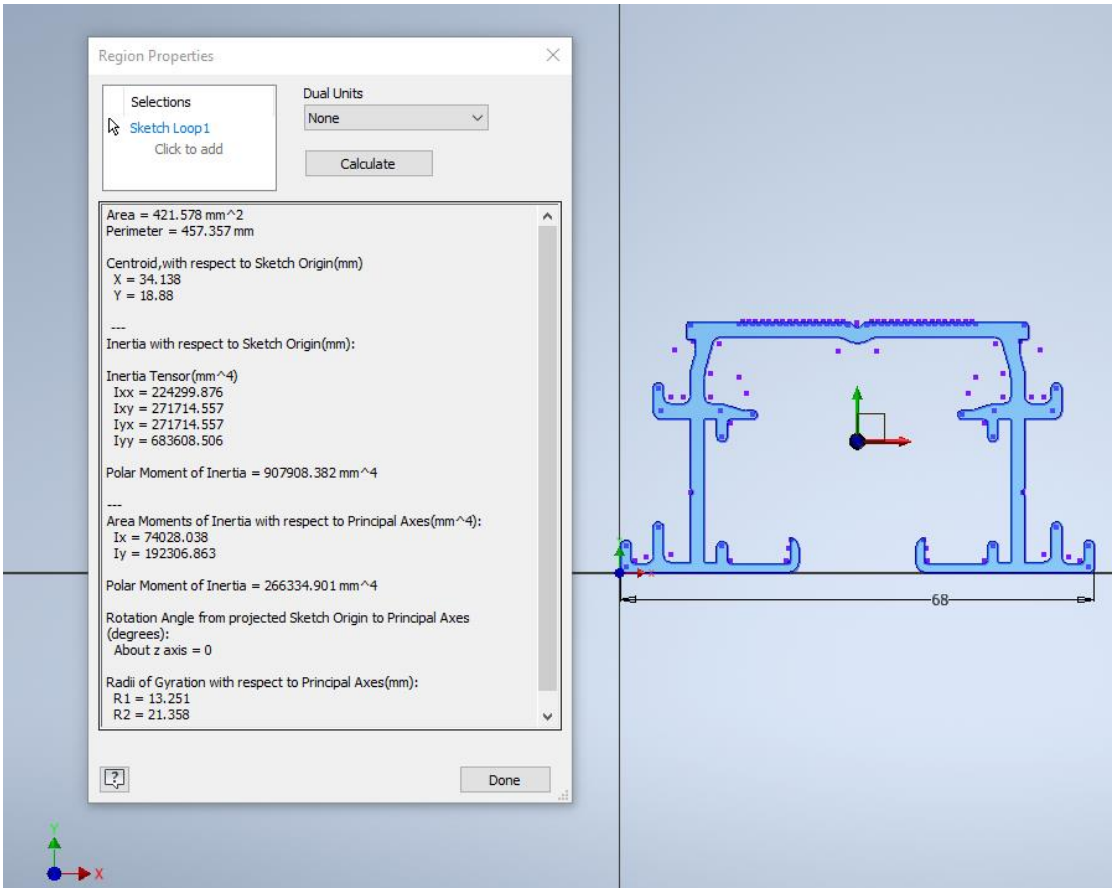
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 74028mm^4$ (35mm Profile K - Series)

$Z = 3920mm^3$ (35mm Profile K - Series)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



35mm Profiles K Series at Every 400mm C/C.

35mm Profile K Series Span is 900mm (Profile Length)

Live Load:

UDL Live load $= 5\text{kN/m}^2 \times 0.4\text{m} = 2\text{kN/m}$ (SLS)

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Project: Aluminium Profile	Contract: 1513-1
Subject: Aluminium Profile Design	Sheet No. 60
Date: 09/04/2021	By: A.N

Dead Load for Composite Deck:

Dead load due to Composite Deck → Convert $\frac{\text{lb}}{\text{m}}$ to $\frac{\text{lb}}{\text{m}^2} \rightarrow \frac{7.3\text{lb/m}}{0.136\text{m}} = 55.15\text{lb/m}^2$

Dead load due to Composite Deck → Convert $\frac{\text{lb}}{\text{m}^2}$ to $\frac{\text{kN}}{\text{m}^2} \rightarrow 55.15\text{lb/m}^2 = 0.245\text{kN/m}^2$

UDL Dead load due to Composite Deck: $0.245\text{kN/m}^2 \times 0.4\text{m} = 0.098\text{kN/m}$ (SLS)

Total UDL Load: $(2\text{kN/m} \times 1.5) + (0.098\text{kN/m} \times 1.35) = 3.13\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{3.13 \times 0.6^2}{8} = 0.141\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{0.141 \times 10^6}{3920} = 36\frac{\text{N}}{\text{mm}^2} < 130\frac{\text{N}}{\text{mm}^2}$$ **Okay**

Maximum Deflection:

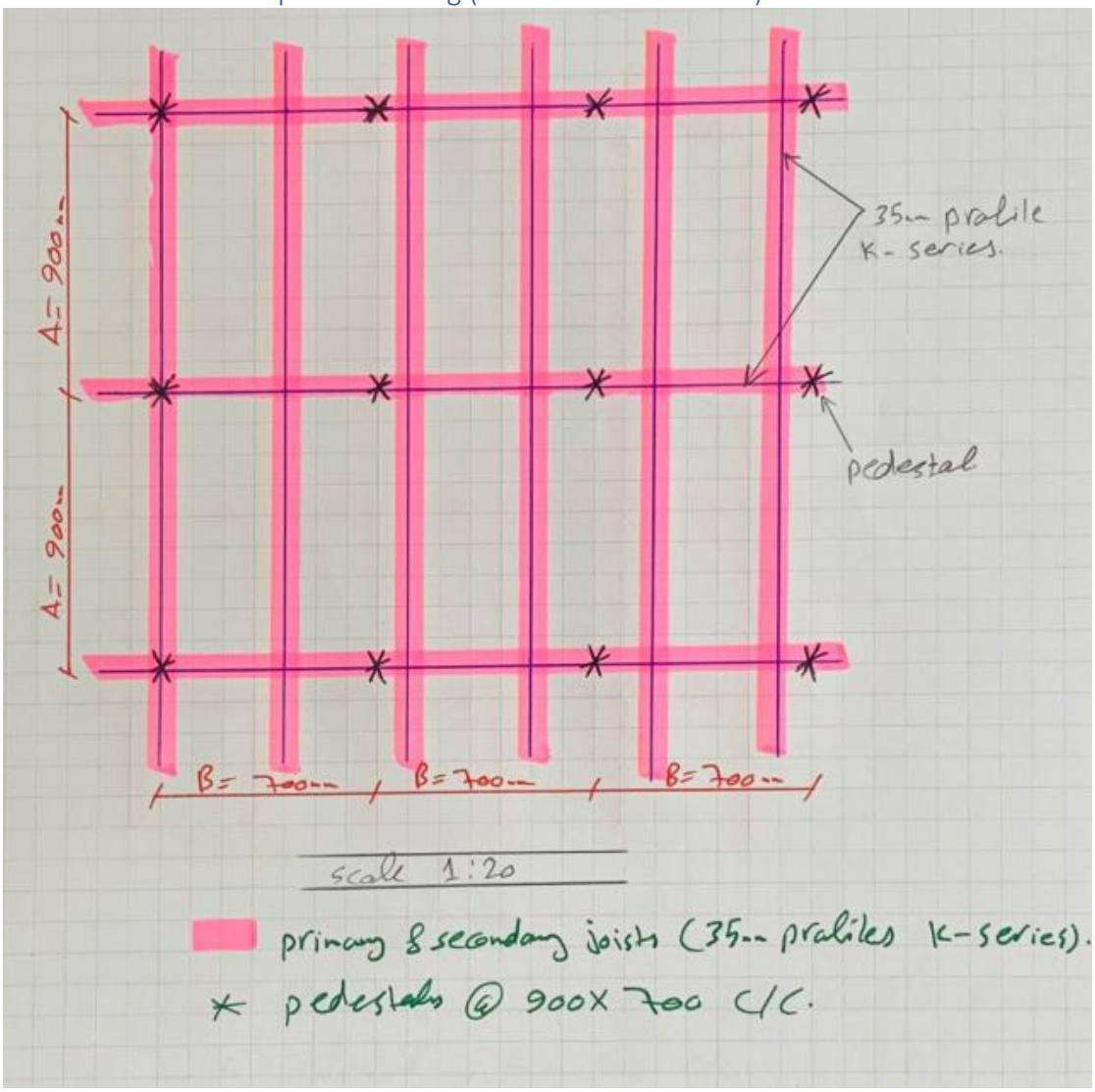
$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 2.098 \times 600^4}{384 \times 70000 \times 74028} = 0.7\text{mm} < 3\text{mm} \rightarrow \left(\frac{600}{200}\right)$$ **Okay**

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 61
Date: 09/04/2021	**By:** A.N

Aluminium Profile Design For the Occasional Case Scenario (35mm Profile K Series):

Sketch Of Grid Line System For the Occasional Case Scenario for 3kN/m2 Live load and Dead load of the Composite Decking (35mm Profile K Series):



Aluminium Profile Checked for Live load of 3kN/m2 and Deal load of the Composite Decking for the Occasional Case Scenario (35mm Profile K Series):

The Profile has the following particulars:

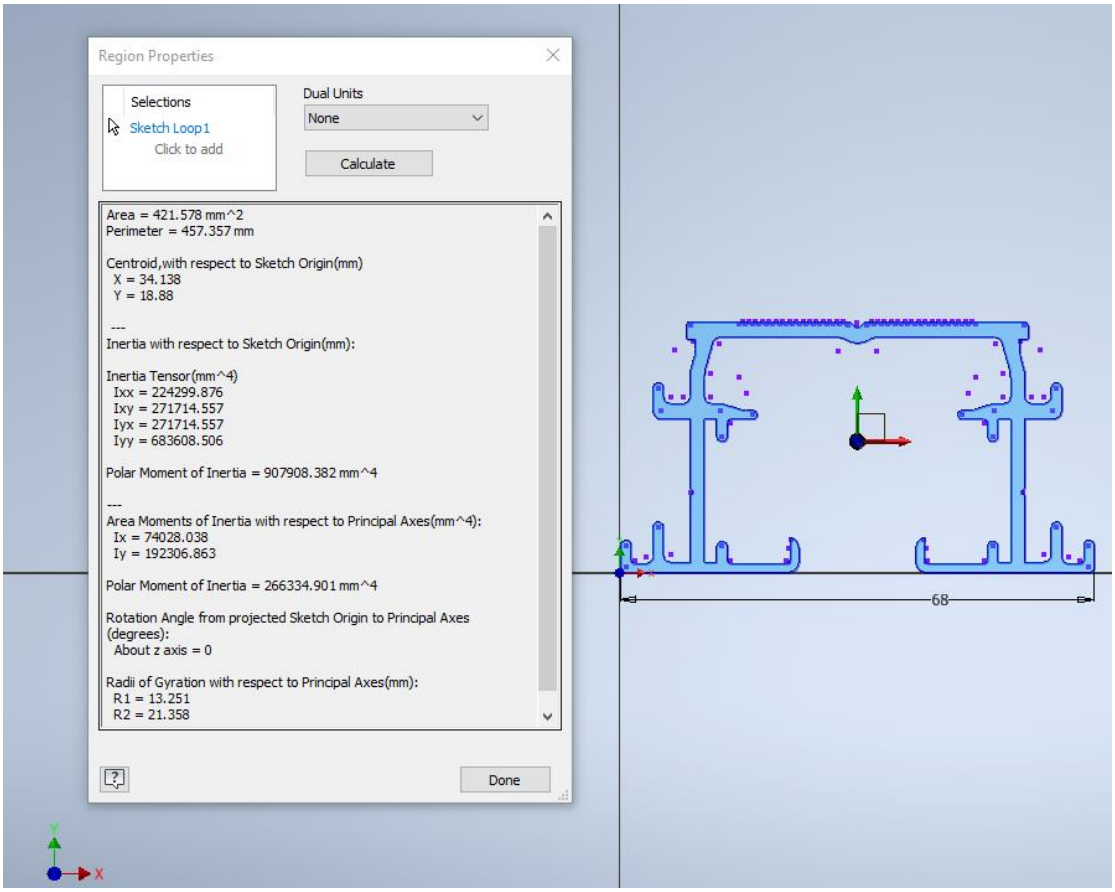
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 74028mm^4$ (35mm Profile K - Series)

$Z = 3920mm^3$ (35mm Profile K - Series)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



Live Load:

UDL Live load $= 3\text{kN/m}^2 \times 0.9\text{m} = 2.7\text{kN/m}$ (SLS)

Dead Load Of Secondary Joists:

Assuming the area of secondary profiles are half of the primary ones:

Dead load due to secondary Joists: $2\text{Nr} \times 0.000457\text{m}^2 \times 27\text{kN/m}^3 = 0.024\text{kN/m}$ (SLS)

Dead Load for Composite Deck:

Dead load due to Composite Deck → Convert $\frac{\text{lb}}{\text{m}}$ to $\frac{\text{lb}}{\text{m}^2} \rightarrow \frac{7.3\text{lb/m}}{0.136\text{m}} = 55.15\text{lb/m}^2$

Dead load due to Composite Deck → Convert $\frac{\text{lb}}{\text{m}^2}$ to $\frac{\text{kN}}{\text{m}^2} \rightarrow 55.15\text{lb/m}^2 = 0.245\text{kN/m}^2$

UDL Dead load due to Composite Deck: $0.245\text{kN/m}^2 \times 0.9\text{m} = 0.22\text{kN/m}$ (SLS)

Total UDL Dead Load: $(0.22 + 0.024\text{kN/m}) = 0.244\text{kN/m}$ (SLS)

Total UDL Load: $(2.7\text{kN/m} \times 1.5) + (0.244\text{kN/m} \times 1.35) = 4.38\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{4.38 \times 0.7^2}{8} = 0.27\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{0.27 \times 10^6}{3920} = 69\frac{\text{N}}{\text{mm}^2} < 130\frac{\text{N}}{\text{mm}^2}$$ **Okay**

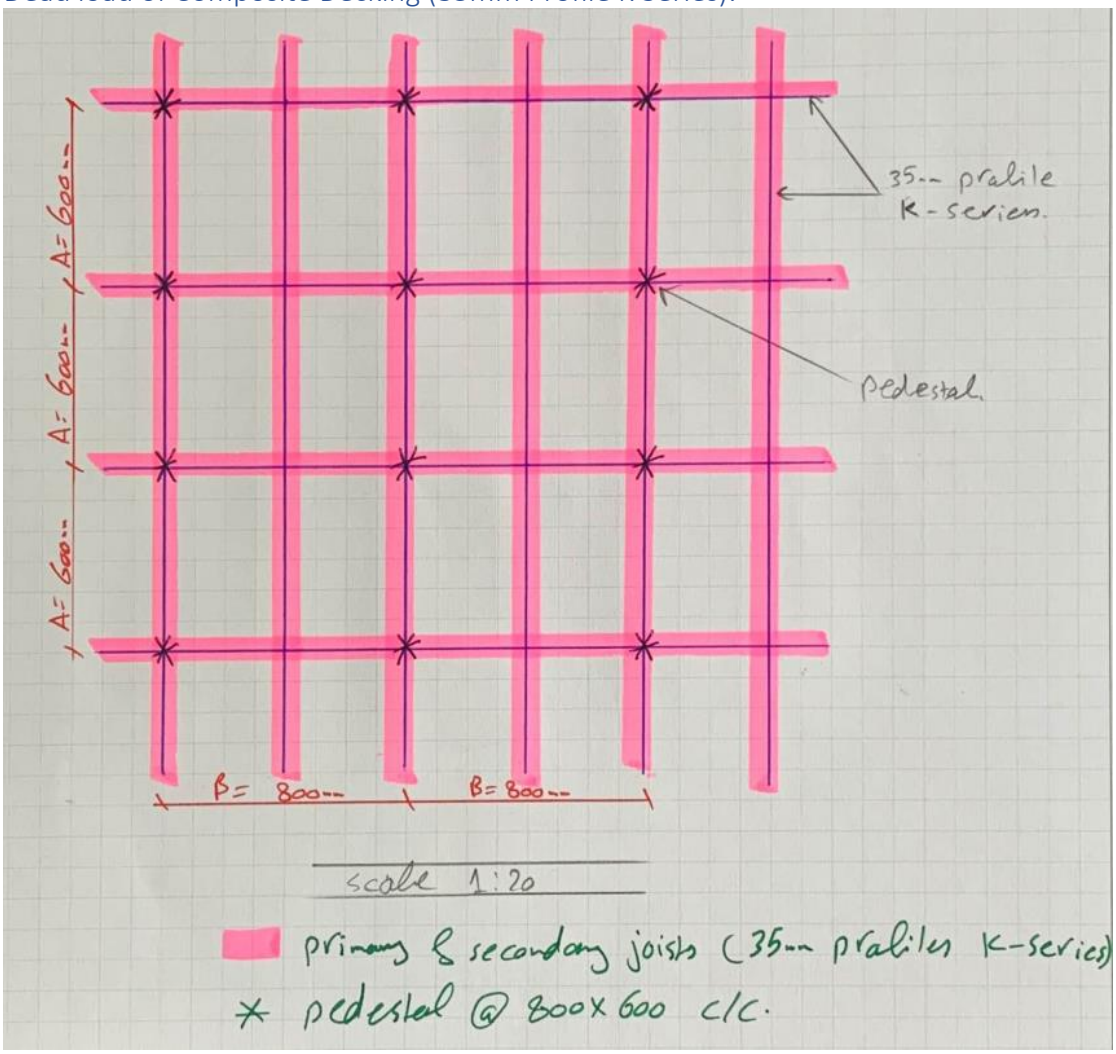
Maximum Deflection:

$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 2.944 \times 700^4}{384 \times 70000 \times 74028} = 1.8\text{mm} < 3.5\text{mm} \rightarrow \left(\frac{700}{200}\right)$$ **Okay**

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 64
Date: 09/04/2021	**By:** A.N

Sketch Of Grid Line System For the Occasional Case Scenario for 5kN/m2 Live load and Dead load of Composite Decking (35mm Profile K Series):



Aluminium Profile Checked for Live load of 5kN/m2 and Dead Load of the Composite Decking for the Occasional Case Scenario (35mm Profile K Series):

The Profile has the following particulars:

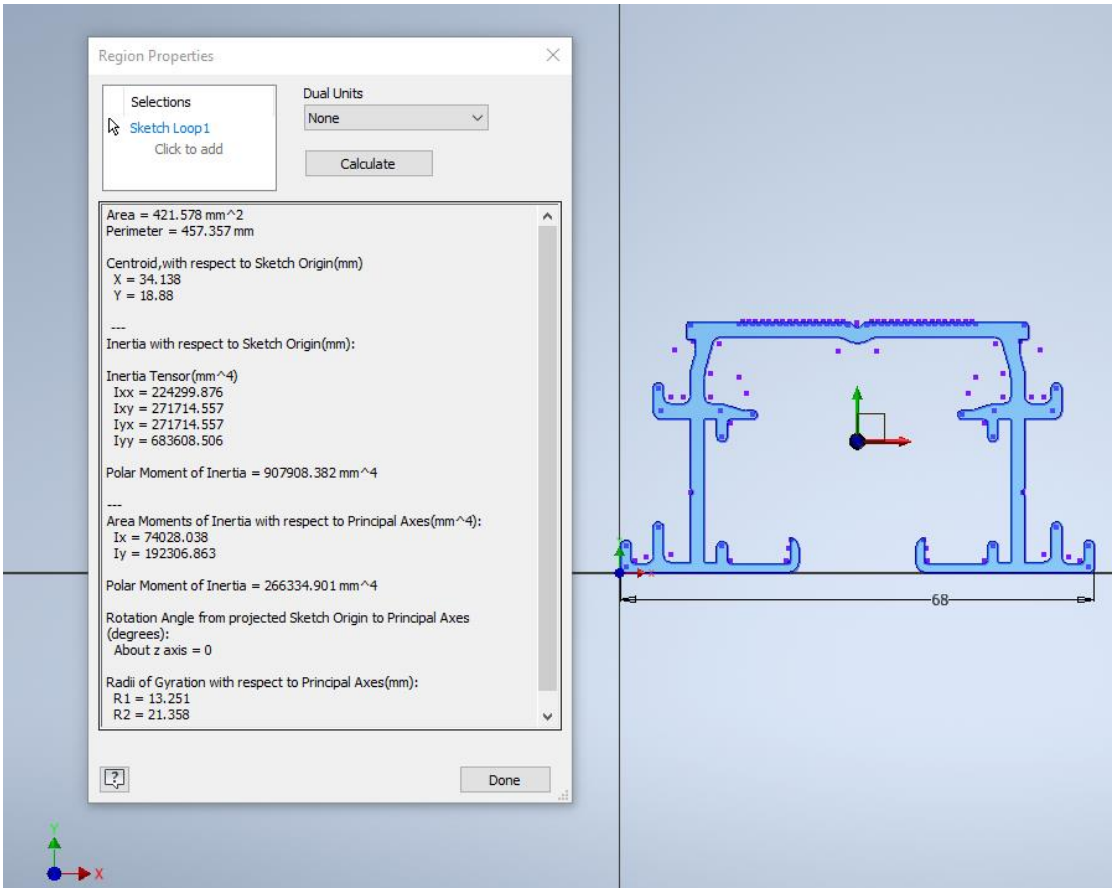
$f_y = 130\ MPa$ (Grade 6063 T-5, Table 3.2b EN 1991-1-1:2007)

$E = 70{,}000\ MPa$ (Grade 6063 T-5, 3.2.5 EN 1991-1-1:2007)

$I = 74028mm^4$ (35mm Profile K - Series)

$Z = 3920mm^3$ (35mm Profile K - Series)

$\gamma_Q = 1.5$ (Table 6.10 EN 1991-1-1:2002)



Live Load:

UDL Live load $= 5\text{kN/m}^2 \times 0.6\text{m} = 3\text{kN/m}$ (SLS)

Dead Load Of Secondary Joists:

Assuming the area of secondary profiles are half of the primary ones:

Dead load due to secondary Joists: $2\text{Nr} \times 0.000457\text{m}^2 \times 27\text{kN/m}^3 = 0.024\text{kN/m}$ (SLS)

Project: Aluminium Profile	**Contract:** 1513-1
Subject: Aluminium Profile Design	**Sheet No.** 66
Date: 09/04/2021	**By:** A.N

Dead Load for Composite Deck:

Dead load due to Composite Deck → Convert $\frac{\text{lb}}{\text{m}}$ to $\frac{\text{lb}}{\text{m}^2} \rightarrow \frac{7.3\text{lb/m}}{0.136\text{m}} = 55.15\text{lb/m}^2$

Dead load due to Composite Deck → Convert $\frac{\text{lb}}{\text{m}^2}$ to $\frac{\text{kN}}{\text{m}^2} \rightarrow 55.15\text{lb/m}^2 = 0.245\text{kN/m}^2$

UDL Dead load due to Composite Deck: $0.245\text{kN/m}^2 \times 0.6\text{m} = 0.147\text{kN/m}$ (SLS)

Total UDL Dead Load: $(0.147 + 0.024\text{kN/m}) = 0.171\text{kN/m}$ (SLS)

Total UDL Load: $(3\text{kN/m} \times 1.5) + (0.171\text{kN/m} \times 1.35) = 4.73\text{kN/m}$ (ULS)

Maximum Moment:

$$M_H = \frac{w \times l^2}{8}$$

$$M_H = \frac{4.73 \times 0.8^2}{8} = 0.378\text{kNm}$$

Maximum Stress:

$$\sigma_{max} = \frac{M}{Z}$$

$$\sigma_{max} = \frac{0.378 \times 10^6}{3920} = 97\frac{\text{N}}{\text{mm}^2} < 130\frac{\text{N}}{\text{mm}^2}$$ **Okay**

Maximum Deflection:

$$\alpha_H = \frac{5wl^4}{384EI}$$

$$\alpha_H = \frac{5 \times 3.171 \times 800^4}{384 \times 70000 \times 74028} = 3.27\text{mm} < 4\text{mm} \rightarrow \left(\frac{800}{200}\right)$$ **Okay**